

GTOS Handbook - Current Project Gap Analysis Appendix

GTOS Enterprise Engineering Handbook - Final Edition 1.0

Source Chapters

- Conversation Intent Review and Access Boundary
- Final Artifact Inventory and Integrity Review
- Content Quality Audit
- Claim and Evidence Audit
- Structure, Numbering, and Canonicalization Audit
- PDF, XLSX, and ZIP Audit
- Critical Findings and Issue Register
- Remediation Decisions
- Repository Evidence Wave 01 - Proforma Invoice
- Repository Evidence Wave 01 - Payment
- Repository Evidence Wave 01 - Purchase Order
- Repository Evidence Wave 01 - Manufacturing
- Evidence Wave 01 Traceability Matrix
- Next Repository Evidence Wave
- Batch 07 Integrity Report
- GTOS Handbook - Batch 07
- 00 — Scope, Method, and Repository Snapshot
- 01 — Quality Inspection, QC Hold, and Release
- 02 — Packing List and Commercial Invoice
- 03 — Export Customs and Legal Release
- 04 — Import Customs, Duties, and Release
- 05 — Transport Booking, Handover, Shipment, and Tracking
- 06 — Warehouse Receiving and Receipt
- 07 — Inventory Synchronization, Reservation, and QC Disposition
- 08 — Cross-Domain Contradictions and Canonicalization Register
- 09 — Implementation Status Matrix

- 10 — Wave 02 Issue Register
- 11 — Wave 02 Traceability Matrix
- 12 — Remediation and Verification Plan
- 13 — Repository Evidence Wave 03 Plan
- GTOS Batch 08 — Repository Evidence Wave 02 Combined Edition
- Implementation and Evidence Status Vocabulary
- Integrity Report
- GTOS Handbook — Batch 08
- Batch 08 Release Report
- 00 — Scope, Method, and Snapshot
- 01 — Product Catalog, Marketplace Catalog, and Offer
- 02 — Pricing, Price Version, and Commercial Snapshot
- 03 — Cart, Checkout, and AI Insights
- 04 — Customer Order and Workflow
- 05 — Allocation, Fulfilment, Pick, Pack, and Last Mile
- 06 — Returns, Inspection, Disputes, and Refund
- 07 — Settlement, Reconciliation, and Payout
- 08 — ESG, Sustainability Goals, and Assurance
- 09 — AI Evaluation and Governed Learning
- 10 — Cross-Domain Conflicts and Canonicalization
- 11 — Implementation Status Matrix
- 12 — Wave 03 Issue Register
- 13 — Wave 03 Traceability Matrix
- 14 — Remediation and Verification Plan
- 15 — Repository Evidence Wave 04 Plan
- GTOS Batch 09 — Repository Evidence Wave 03 Combined Edition
- Implementation and Evidence Status Vocabulary
- Integrity Report
- GTOS Handbook — Batch 09
- Batch 09 Release Report

Conversation Intent Review and Access Boundary

Access statement

The platform does not expose a byte-for-byte export of the entire conversation transcript, hidden tool payloads, or private reasoning as a file. Therefore this review does not falsely claim a literal line-by-line export from the first UI character.

The review used:

- all user and assistant messages available in the active conversation context;
- the safe conversation summary available to the assistant;
- all generated conversation artifacts visible in /mnt/data;
- all final ZIPs, Markdown files, PDF, XLSX catalogue, manifests, release reports, and GitHub release markers available in this session.

Reconstructed non-negotiable intent

- The Handbook must be written in English; progress updates may be in Persian.
- The work documents the project; it does not fix application code.
- Repository executable source, migrations, runtime configuration, tests, and deployment evidence outrank plans and READMEs.
- Reality != Knowledge about Reality is foundational.
- Recommendation, Decision, Intent, execution, and outcome remain separate.
- The three-Portal lifecycle and the complete global-trade flow must be covered.
- AI Orchestrator, AI Runtime, MCP, Workflow Engine, Event Bus, and Domain ownership must not be collapsed.
- AI must not write directly to Domain databases or create shadow truth.
- Every chapter should cover ownership, state, rules, Commands, Events, APIs, data, Portals, AI, security, reliability, tests, operations, examples, and repository evidence.
- Unverified behavior must never be presented as implemented.
- The final corpus should be exhaustive, but page count is not the completion criterion.

Review conclusion

The direction and conceptual hierarchy remain correct. The content-maturity claims of Batches 02-06 were not correct and have been superseded by corrected AUDITED-v2 packages.

Final Artifact Inventory and Integrity Review

Audited artifact families

- prior consolidated compilation ZIP;
- 266-page compiled PDF;
- XLSX artifact catalogue;
- Batch 01 through Batch 06 ZIPs and extracted directories;
- original Master ZIP;
- corrected AUDITED-v2 Batch ZIPs and corrected Master;
- manifests, release reports, integrity reports, scripts, and GitHub markers.

Batches 01-06 content inventory

- Chapter Markdown files audited: **620**
- Words in audited chapter corpus: **1,245,335**
- Empty chapter files: **0**
- Exact duplicate-file groups: **0**
- Original Master ZIP members: **36**
- Nested Batch ZIPs in original Master: **6**
- Master CRC error: **None**

Manifest and archive results

All inspected Batch ZIPs passed CRC validation. Manifest paths and SHA-256 values matched the extracted files at audit time. The original Master is a packaging container of nested ZIPs and reports; it is not an expanded source tree.

Prior consolidated compilation

- Files: **591**
- Markdown source files: **258**
- Markdown words: **42,939**
- Median words per Markdown file: **103.5**
- Broken internal Markdown links: **0**
- Empty files: **0**

The prior package is structurally useful and link-clean, but it is a compact reference edition rather than the requested repository-grounded comprehensive Handbook.

Content Quality Audit

Corpus-wide result

The generated Batches 02-06 contain a large amount of repeated architecture-template text.

- Common-line instance ratio across Batches 01-06: **80.2%**
- Batch 01 mean common-line ratio: **22.3%**
- Batch 02 mean common-line ratio: **82.5%**
- Batch 03 mean common-line ratio: **83.0%**
- Batch 04 mean common-line ratio: **81.5%**
- Batch 05 mean common-line ratio: **83.2%**
- Batch 06 mean common-line ratio: **83.1%**

Interpretation

Batch 01 contains more stage-specific material, although it still requires source enrichment. Batches 02-06 are primarily architecture skeletons generated from one common chapter pattern. The chapter titles, owners, purposes, and a small number of phrases vary; most lifecycle, API, table, security, SLO, testing, and checklist sections are repeated.

Maturity correction

The repeated status Draft Complete for Architecture Review was too strong. It has been replaced in AUDITED-v2 packages by:

Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required

and:

TEMPLATE_DERIVED_REFERENCE_SKELETON

Quality decision

The skeletons may be retained as checklists and target-architecture scaffolding. They must not be counted as completed repository specifications or as evidence that the 5,000-8,000 page objective has been achieved.

Claim and Evidence Audit

Concrete evidence found in the generated chapter corpus

Evidence kind	Count
concrete backend/ paths	1
concrete frontend paths	0
java implementation filenames	0
migration SQL filenames	0
commit SHAs	0
source line citations	0
concrete service ports	0

The thousands of API-route matches are generated reference surfaces repeated across chapters, not inspected repository routes.

Critical distinction

Reference API proposed by the Handbook

!= route found in source

!= behavior verified by test

!= behavior observed at runtime

Required upgrade evidence

A chapter may be upgraded from `TEMPLATE_DERIVED_REFERENCE_SKELETON` only after it identifies, at minimum:

- exact repository paths and verified commit;
- entities, aggregates, state fields, and lifecycle methods;
- controllers, route methods, request/response types, and authorization;
- migrations, table names, keys, indexes, and ownership;
- Events, outbox, consumers, replay, and workflow links;
- Portal routes and client contracts;
- tests proving positive, negative, concurrency, tenant, security, and recovery behavior;
- deployment and operational evidence where relevant;
- specific gaps and contradictions.

Structure, Numbering, and Canonicalization Audit

Findings

1. The canonical lifecycle contains 27 stages, while the publication contains 28 numbered stage chapters because invitation/submission and negotiation/selection were split.
2. The publication-to-canonical map created in Batch 01 is necessary and remains valid as a transitional mapping.
3. Earlier Reference Theory and Canon directories contain alternate or duplicate filenames that require a canonical index and deprecation redirects.
4. Purchase Order, RFQ, Quotation, Production Order, AI/MCP, service catalogues, ports, and database ownership contain repository-level contradictions.
5. Volume-level chapter counts are not completion evidence.

Decision

- Preserve canonical lifecycle IDs independently from publication chapter numbers.
- Preserve duplicate source artifacts until a canonicalization decision records their disposition.
- Add `canonical_owner`, `implementation_owner`, `compatibility_owner`, and `retirement_status` to registries.
- Never merge similarly named concepts solely by filename similarity.

PDF, XLSX, and ZIP Audit

PDF

The compiled prior PDF has 266 A4 pages, embedded Unicode fonts, a large outline, and no encryption. Rendered samples from the first, middle, and final pages showed readable text, working tables, and no observed clipping or broken glyphs.

The PDF represents the earlier compact source edition plus the conversation PI draft. It does not contain the new Batch 01-06 corpus and must not be described as the final expanded Handbook.

XLSX catalogue

The workbook opens correctly and contains Summary, Local Files, GitHub Branch, Status Vocabulary, and Releases sheets with no detected formula errors.

However, its classification of 130 GitHub paths as `EMPTY_REPOSITORY_ARTIFACT` was based on zero additions from a compare response. Direct file reads later proved that some of those files contain substantial content. The workbook is therefore structurally valid but semantically superseded for branch-content classification.

ZIPs

All inspected ZIPs passed CRC checks. The original Master ZIP contains nested Batch ZIPs and reports, not an expanded merged source tree. The corrected Master AUDITED-v2 contains the corrected Batch packages and audit JSON files.

Critical Findings and Issue Register

ID	Severity	Finding	Required action
AUD-001	Critical	Batches 02-06 were labelled Draft Complete despite being template-derived	use Audited-v2 packages and expand subject-specific content
AUD-002	Critical	repository evidence in generated corpus is near zero	perform code-by-code evidence enrichment
AUD-003	Critical	word and file counts were used in ways that could imply completion	use semantic completion gates instead
AUD-004	High	original Master is nested packaging, not merged source	publish an expanded canonical source tree later
AUD-005	High	XLSX zero-addition classification is unreliable	rebuild catalogue from direct file reads
AUD-006	High	canonical lifecycle and publication numbering differ	retain explicit mapping and stable IDs
AUD-007	High	generic reference APIs and tables are repeated as if chapter-specific	replace with verified or clearly target-specific contracts
AUD-008	High	major Domain ownership conflicts remain	create canonicalization ADRs and ownership matrices
AUD-009	Medium	prior compact edition is only 42,939 Markdown words across 258 files	retain as legacy reference, not final edition
AUD-010	Medium	no current PDF exists for the expanded corpus	defer PDF until content review and deduplication

Gate

No new mass template generation is permitted until the current subject family is enriched with concrete repository evidence and reviewed for ownership, duplication, and correctness.

Remediation Decisions

Completed in this Batch

- audited all available final Batch Markdown files line by line programmatically;
- verified ZIP CRC and Manifest hashes;
- inspected the prior PDF visually and structurally;
- inspected the XLSX workbook structure and formulas;
- created corrected AUDITED-v2 packages for Batches 01-06;
- changed overstated maturity labels;
- created corrected Master AUDITED-v2;
- began repository evidence enrichment for PI, Payment, PO, and Manufacturing.

Mandatory next actions

1. Enrich one Domain or lifecycle area at a time from executable evidence.
2. Calculate a chapter evidence score and template-residual score.
3. Replace generic aggregate/API/table sections with repository-specific and target-specific sections.
4. Resolve ownership conflicts through explicit ADRs.
5. Add exact tests, migrations, routes, schemas, Portals, and operational evidence.
6. Regenerate catalogues from direct source reads.
7. Build PDF only after deduplication and content review.

Repository Evidence Wave 01 - Proforma Invoice

Evidence baseline

Verified commit: 14cc83805e7386c3d5cdec6ad191b1078fd502ff.

Inspected files:

- backend/domain/commerce/tajerestan-services/src/main/java/com/allinb2c/commerce/tajerestan/entity/ProformaInvoiceEntity.java
- backend/domain/commerce/tajerestan-services/src/main/java/com/allinb2c/commerce/tajerestan/service/ProformaInvoiceService.java
- backend/domain/commerce/tajerestan-services/src/main/java/com/allinb2c/commerce/tajerestan/api/ProformaInvoiceController.java

Verified implementation

ProformaInvoiceEntity maps to tb_proforma_invoice. It records PI number, supplier, customer, quotation and purchase-order references, Incoterm, currency, payment terms, prepayment ratio, ports, country, shipping method, estimated ship date, expiry date, total amount, status, notes, company, timestamps, items, and payment schedules.

The documented status vocabulary in source is DRAFT, SENT, ACCEPTED, CONVERTED, and CANCELLED.

ProformaInvoiceService implements:

- create PI and line items;
- get and list;
- send;
- accept;
- convert to a Purchase Order reference;
- Events trade.proforma-invoice.created, .sent, .accepted, and .converted.

ProformaInvoiceController exposes:

- POST /api/v1/trade/proforma-invoices;
- GET /api/v1/trade/proforma-invoices/{id};
- GET /api/v1/trade/proforma-invoices;
- PUT /api/v1/trade/proforma-invoices/{id}/send;
- PUT /api/v1/trade/proforma-invoices/{id}/accept;
- POST /api/v1/trade/proforma-invoices/{id}/convert.

Verified gaps

- Service uses a constant DEFAULT_COMPANY_ID and also assigns it as customer ID.
- supplierId can become null through safeUUID, while the Entity declares it non-null.
- No immutable PI version entity or optimistic-lock field was found.
- Send, accept, and convert methods do not enforce explicit source-state guards.
- The purchaseOrderId parameter accepted by list service is not used in the inspected query selection.
- PI number generation is time/random based and not a governed numbering policy.
- The inspected code does not establish approval, signature, beneficiary verification, expiry enforcement, or payment-risk approval.

Status decision

Capability	Status
PI entity and persistence mapping	VERIFIED_IMPLEMENTED

Capability	Status
create/get/list/send/accept/convert endpoints	VERIFIED_IMPLEMENTED
line items and payment-schedule relationships	VERIFIED_IMPLEMENTED
basic trade Events	VERIFIED_IMPLEMENTED
tenant-context enforcement	IMPLEMENTED_WITH_GAP
lifecycle guards	IMPLEMENTED_WITH_GAP
immutable commercial versioning	PLANNED_NOT_IMPLEMENTED in inspected source
formal approval/signature/risk workflow	REQUIRES_REPOSITORY_REVIEW

Repository Evidence Wave 01 - Payment

Evidence baseline

Inspected files:

- backend/domain/commerce/payment-service/src/main/java/com/allinb2c/commerce/payment/entity/PaymentIntentEntity.java
- backend/domain/commerce/payment-service/src/main/java/com/allinb2c/commerce/payment/service/PaymentIntentService.java
- backend/domain/commerce/payment-service/src/main/java/com/allinb2c/commerce/payment/api/PaymentController.java

Verified implementation

PaymentIntentEntity maps to payment_service_intents and defines indexes for order, status, provider, customer, creation, expiry, and tenant. It records amount, currency, capture mode, status, provider, provider reference, customer, description, expiry, errors, metadata, a unique idempotency key, and a JPA optimistic-lock version.

Lifecycle methods support captured, partially captured, cancelled, expired, and failed states. Captured intents cannot be cancelled.

PaymentIntentService implements creation, idempotency lookup, tenant-aware queries, pagination, provider/customer/status filters, and cancellation. The service obtains tenant context from a gateway-injected header and uses unscoped behavior when the context is absent for internal flows.

PaymentController exposes payment intents, capture, refund, settlement, payout, webhook, dispute, summary, and provider information under /api/payments.

Verified gaps and risks

- Legacy/internal unscoped access is deliberately preserved and requires strong trusted-caller controls.
- Tenant context depends on a gateway-injected header; direct-service access must be prevented or independently authenticated.
- Idempotency lookup is by idempotency key without an inspected tenant predicate; a global unique key may cause cross-tenant behavior unless keys are globally namespaced and responses are protected.
- The inspected payment-intent status model starts at REQUIRES_CAPTURE; separate authorization, risk approval, and Outcome Unknown states are not part of this Entity.
- No formal PI risk-approval Decision aggregate was established in the inspected files.
- Webhook signature verification and provider-outcome reconciliation require inspection of additional classes.

Status decision

Capability	Status
payment-intent entity, indexes, expiry and lifecycle methods	VERIFIED_IMPLEMENTED
idempotency key and optimistic locking	VERIFIED_IMPLEMENTED
tenant-aware intent reads	VERIFIED_IMPLEMENTED with legacy unscoped path
intent/capture/refund/settlement/payout/webhook API surface	VERIFIED_IMPLEMENTED
trusted gateway and direct-access enforcement	REQUIRES_REPOSITORY_REVIEW
PI payment-risk approval workflow	PARTIAL_IMPLEMENTATION / separate capability not verified
provider signature and Outcome Unknown reconciliation	REQUIRES_REPOSITORY_REVIEW

Repository Evidence Wave 01 - Purchase Order

Evidence baseline

Two independent implementations were inspected.

Procurement:

- backend/domain/commerce/procurement-service/.../PurchaseOrderEntity.java
- backend/domain/commerce/procurement-service/.../PurchaseOrderService.java
- backend/domain/commerce/procurement-service/.../PurchaseOrderController.java

Tajerestan:

- backend/domain/commerce/tajerestan-services/.../PurchaseOrderEntity.java
- backend/domain/commerce/tajerestan-services/.../PurchaseOrderService.java

Procurement implementation

The Procurement Entity maps to `procurement.purchase_orders` and records tenant, company, PO number, supplier, RFQ, status, order and delivery dates, currency, financial totals, notes, creator, approver, and timestamps.

The service implements list by tenant/company, get, create, full update, delete, and item CRUD. The controller exposes these operations under `/api/procurement/purchase-orders`.

Tajerestan implementation

The Tajerestan Entity maps to an unqualified `purchase_orders` table. It includes RFQ and quotation references, buyer and supplier identities/names, Incoterm, shipping, tax and totals, balance due, payment terms, tracking, carrier, planned/actual dates, approval, cancellation, completion, customs fields, documents, tenant, and line items.

The service implements creation and financial calculation, list/search, update, approve, receive goods, statistics, and Events `trade.po.created`, `.updated`, `.approved`, and `.goods-received`.

Critical ownership conflict

The repository contains at least two materially different Purchase Order aggregates with different packages, schemas, identity types, fields, APIs, lifecycle scope, and responsibilities. Neither can be silently declared the canonical PO without an ownership decision.

Verified gaps

- Procurement get-by-ID is not tenant-scoped in the inspected service.
- Procurement accepts tenant/company from request data and its controller exposes tenant as query input; trusted-context enforcement was not found in the inspected files.
- Procurement full update can replace status and commercial values without state guards or immutable versions.
- Procurement delete physically removes items and the PO, conflicting with historical-integrity requirements.
- Tajerestan update and approve methods use direct string state mutation without visible transition guards.
- Tajerestan `receiveGoods` mixes warehouse receiving and PO state.
- Supplier acknowledgement is not represented as a distinct aggregate or Decision in inspected files.

Status decision

Capability	Status
Procurement PO CRUD and item operations	VERIFIED_IMPLEMENTED
Tajerestan PO creation, approval, receipt and Events	VERIFIED_IMPLEMENTED
canonical PO ownership	CONFLICT_REQUIRES_CANONICALIZATION
tenant isolation	IMPLEMENTED_WITH_GAP / REQUIRES_REPOSITORY_REVIEW
immutable PO versions and governed amendments	PLANNED_NOT_IMPLEMENTED in inspected code
supplier acknowledgement	REQUIRES_REPOSITORY_REVIEW
historical integrity	IMPLEMENTED_WITH_GAP because destructive delete exists

Repository Evidence Wave 01 - Manufacturing

Evidence baseline

Inspected files:

- backend/domain/commerce/manufacturing-service/src/main/java/com/allinb2c/commerce/manufacturing/model/entity/ProductionOrderEntity.java
- backend/domain/commerce/manufacturing-service/src/main/java/com/allinb2c/commerce/manufacturing/service/ProductionOrderService.java
- backend/domain/commerce/manufacturing-service/src/main/java/com/allinb2c/commerce/manufacturing/controller/ProductionOrderController.java

Search also found a second ProductionOrderEntity in warehouse-service, requiring ownership review.

Verified implementation

The manufacturing Entity maps to manufacturing.tb_production_order. It records production-order number, purchase order, supplier, product, quantity, unit, planned and actual dates, status, priority, notes, company, and timestamps.

The service implements:

- production-order list/get/create/update/delete;
- production materials with required, purchased, and received quantities;
- production progress with stage, percentage, completed and defective quantity, images, videos, reporter, and time;
- production schedules with line, date, shift, planned output, actual output, and status.

The controller exposes production-order, material, progress, and schedule routes under /api/manufacturing/production-orders.

Verified gaps

- No tenant field exists on the inspected production-order Entity; company ID is supplied in requests.
- Controller and service methods shown do not establish authorization or company-scope validation.
- Full update permits arbitrary status and field replacement without transition guards.
- Delete removes materials, progress, schedules, and the order, conflicting with history-preservation principles.
- No optimistic lock/version field was found.
- No Events or transactional outbox were found in the inspected service.
- Capacity commitment, factory qualification, BOM/routing, work orders, lot genealogy, quality hold points, and immutable baseline/rebaseline are not modeled in the inspected files.
- A second Production Order implementation under Warehouse indicates an ownership conflict.

Status decision

Capability	Status
production-order persistence and CRUD	VERIFIED_IMPLEMENTED
material, progress and schedule records	VERIFIED_IMPLEMENTED
Portal/API route surface	VERIFIED_IMPLEMENTED
tenant and authority enforcement	REQUIRES_REPOSITORY_REVIEW
history preservation and lifecycle guards	IMPLEMENTED_WITH_GAP
manufacturing Events/outbox	PLANNED_NOT_IMPLEMENTED in inspected service
capacity and immutable production planning	PARTIAL_IMPLEMENTATION
canonical Production Order ownership	CONFLICT_REQUIRES_CANONICALIZATION

Evidence Wave 01 Traceability Matrix

Area	Entity/Table	Service	Controller/API	Events	Principal status
Proforma Invoice	tb_proforma_invoice, items, schedules	ProformaInvoiceService	/api/v1/trade/proforma-invoices	created/sent/accepted/converted	IMPLEMENTED_WITH_GAP
Payment	payment_service_intents and related payment tables	PaymentIntentService / application service	/api/payments	requires additional Event inspection	IMPLEMENTED_WITH_GAP
Purchase Order - Procurement	procurement.purchase_orders and items	Procurement PurchaseOrder-Service	/api/procurement/purchase-orders	none found in inspected service	IMPLEMENTED_WITH_GAP
Purchase Order - Tajerestan	purchase_orders and line items	Tajerestan PurchaseOrder-Service	Tajerestan API requires separate route profile	trade PO Events	CONFLICT_REQUIRES_CANONICALIZATION
Manufacturing	manufacturing.tb_production_order, materials, progress, schedules	ProductionOrderService	/api/manufacturing/production-orders	none found in inspected service	PARTIAL_IMPLEMENTATION

Next evidence targets

- migrations and indexes for all four areas;
- security configuration and method/route authorization;
- tests, tenant isolation, concurrency, and negative transitions;
- Portal pages and client contracts;
- EventPublisher implementation, topics, outbox, consumers, replay, and DLQ;
- provider/webhook verification and reconciliation;
- canonical ownership ADRs for PO and Production Order.

Next Repository Evidence Wave

Wave 02 scope

1. Quality inspection and release;
2. packing list and commercial invoice;
3. export/import customs;
4. shipping booking, tracking, and custody;
5. warehouse receiving and inventory synchronization.

Wave method

For each area:

- inspect entity, repository, service, controller, DTO, migration, tests, frontend routes, deployment, and Events;
- produce an implementation map;
- classify each capability;
- identify ownership conflicts and destructive-history patterns;
- rewrite the corresponding skeleton chapters;
- update traceability and the direct-read catalogue;
- commit code-grounded chapters to the Handbook branch.

Batch 07 Integrity Report

- Markdown files: 15
- Empty Markdown files: 0
- Manifest entries: 19
- Audited chapter corpus: 620 files
- Corrected Master included: True
- Result: PASS

GTOS Handbook - Batch 07

Audit, Canonicalization, and Repository Evidence Enrichment Wave 01

Batch 07 is a corrective release. It does not add another large template corpus. It audits every available final artifact, corrects maturity labels, records critical issues, and upgrades four high-priority areas with concrete repository evidence:

1. Proforma Invoice;
2. Payment Intent and execution surface;
3. Purchase Order ownership conflict;
4. Manufacturing production orders, materials, progress, and schedules.

The previous Batches 01-06 remain historical artifacts. Their AUDITED-v2 versions supersede them for review and planning decisions.

00 — Scope, Method, and Repository Snapshot

Field	Value
Repository	chinair88-prog/allinb2c
Evidence snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Handbook branch	gtos/handbook-repository-grounded-expansion-v2
Wave	Repository Evidence Wave 02
Scope	Quality: documents: customs: transport: warehouse: inventory
Evidence standard	شده بررسی اجرای فایل با شده پشتی بانی و محدود ادعای

بررسی روش

فایل نام وجود شدند. خواننده مستقیماً SQL و JavaScript و Java، منتخب های فایل سپس و شد انجام کد و جوی جست ابتدا نشد. محسوب Implementation تنه ایی به route ی

اند: شده حفظ تمایزها این ها فصل تمام در

1. frontend presentation در برابر backend execution؛
2. software quality در برابر trade-goods quality؛
3. platform status در برابر external legal authority؛
4. کامل؛ lifecycle برابر در جدول وجود
5. duplicate؛ از واقعی جلوگیری برابر در idempotency key ذخیره
6. اعتماد؛ قابل tenant enforcement برابر در tenant field وجود
7. شده؛ کنترل state machine برابر در status string
8. outbox. و شده تضمین delivery برابر در Event publisher فراخوانی

هم محدودیت

- Deployment این snapshot اثبات
- نیست. CI یا Test موفق اجرای معنی به Test فایل وجود
- migration؛ اجرای نه اند، شده بررسی source روی از migration؛
- outcome؛ customs, carrier, inspector و warehouse با authority evidence توان ندمی
- است. integration test نیازمند migration با runtime datasource هماهنگی و دارند JDBC direct controller؛ از بخشی

01 — Quality Inspection, QC Hold, and Release

اجرای نتایج

م تفاوت مفهوم سه نشد. پیدا شده بررسی شواهد در تجاری کالای کیفی بازرسی برای canonical و مستقل Aggregate اند: شده مخلوط Quality وازه با

1. می محلی ثابت آرایه یک از را Inspection های داده apps/quality-dashboard/app/inspections/page.tsx صفحه 1. خواند.
2. می مدیریت را report و gate metric، افزار/پروژه، نرم کیفی در dashboard-service در QualityController
3. دارند. stock disposition و QC Hold محدودتر فرآیند یک Warehouse و Inventory

Capability	Status
Trade inspection planning and evidence	REFERENCE_ARCHITECTURE
Quality Dashboard inspection page	FRONTEND MOCK
Software-quality gates/reports	VERIFIED_IMPLEMENTED، اما Domain تفاوت
Stock QC hold and disposition	PARTIAL_IMPLEMENTATION
Final release certificate	REQUIRES_REPOSITORY_REVIEW

Inspection صفحه

نمایش را pending و fail و pass های نتیجه و outgoing و final، in-process، incoming، های نوع Quality Dashboard صفحه شوند. می اجرا آرایه همان روی فقط نیز آمار و وجو است فیلتر، اند. شده تعریف INSPECTIONS ثابت در رکوردها اما ده، می شود. معرفی backend inspection اثبات عنوان به نباید صفحه این

Software Quality Controller

دارد: را زیر رفتارهای /api/quality مسیری

- dashboard و score بر اساس projectId:
- trend های metric:
- ingestion و aggregation metric:
- check و status و failed conditions برای quality gate:
- export و schedule report، تولدی.

این Production Lot، Sample، Measurement، Specification، Inspector، Laboratory، Defect و Release Certificate فاقد مدل این نیست. کالای کیفی Stage به متعلق بنابراین است؛

Inventory QC واقعی رفتار

کند: می سازی پیاده را زیر رفتارهای InventoryPostingService

- QC Hold به Available از quantity انتقال:
- available quantity بودن کافی کنترل:
- Available به بازگشت و PASS نوع disposition:
- Damaged به انتقال و FAIL_DAMAGED:
- Blocked به انتقال و FAIL_BLOCKED:
- QC_HOLD، QC_PASS، QC_FAIL_DAMAGED و QC_FAIL_BLOCKED های ledger event ثبت

specification و sampling plan چه کرده، بازرسی کسی چه کنند می ثابت اما کنند، می ثابت را stock disposition رفتار این است. کرده صادر را release ای authority چه و شده ثبت های measurement چه شده، استفاده

QC Hold Workbench

Warehouse UI:

- کند؛ می است فاده `inventoryApi.listQcHolds()` از
- کند؛ می فراخوانی `inspect/release/reject` ای action با را `PATCH /api/proxy/quality/status`
- دارد؛ `PENDING`, `INSPECTING`, `RELEASED` و `REJECTED` ای state
- شود. می `Stock Adjustment` باعث `Reject` و `Inventory Restock` باعث `Release` کند می ادعا

backend متناظر `/quality/status` در `Wave` این شواهد این `CLIENT_CONTRACT_UN-VERIFIED` side effect! این بنابر این نشد؛ پیدا `Wave` این شواهد در `/quality/status` متناظر backend هستند.

مفرد Subjects

اند: لازم Subjects این هنوز کیفیت Stage تکمیل برای

- `InspectionPlan`
- `Inspection`
- `Sample`
- `Measurement`
- `SpecificationVersion`
- `Defect`
- `Nonconformance`
- `CorrectiveAction`
- `QualityDecision`
- `ReleaseCertificate`
- `ReleaseRevocation`
- `Inspector/Laboratory mandate` و `calibration evidence`

اصلاحی اقدامات

1. باشد. داشته جدا `ownership` و گذاری نام `Trade Quality` از `Software Quality`
2. شود. گذاری عملیات `Demo` صریحاً یا متصل واقعی API به `Mock inspection UI`
3. شود. ایجاد `Inventory` از مسئولیت `Quality Domain`
4. شود. متصل `Inventory disposition` به شده تایی `Command` طریقی از `Quality Decision`
5. شوند. ثبت `revocation` و `rationale`, `evidence`, `inspector identity`, `specification version`
6. شود. `integration-test` و مسئولیت پیدا، `/quality/status` به مربوط `proxy`

02 — Packing List and Commercial Invoice

Ownership

دارد: وجود Packing List مدل دو

- `Tajerestan` با `JPA` و جدول `tb_packing_list`؛
- `Provider/Partner` با `direct JDBC` و جدول `partner.packing_lists`.

شود. می نگه‌داری `partner.commercial_invoices` در نیز `Commercial Invoice`

است. `CONFLICT_REQUIRES_CANONICALIZATION` برابر `Packing List` وضعیت، `System of Record` تعین زمان تا

Tajerestan Packing List

فیلدهاست: این شامل واقعی Entity

- `PL` شماره؛
- `Purchase Order reference` و `PI`؛
- `supplier`؛
- `box count`؛
- `net/gross weight` و `volume`؛
- `packing method` و `marks`؛
- `status` و `company`؛
- `child boxes`.

واقعی: Service

- `PL` - شماره می تولدی . . .
- `package count`, `gross weight` و `volume` را جمع می کند؛
- `box` را ذخیره می کند؛
- `list` و `get` دارد؛
- `trade.packing-list.created` نام با Event می منتشر

Gapهای

- `DEFAULT_COMPANY_ID` می است فاده `company` هم و `supplier` برای هم
- `totalNetWeight` می صفر
- `netWeightKg` هر `box` صفر
- `immutable issued version`, `signature` و `supersession` می نشد.
- `status transition guard` و `optimistic locking` می نشد.
- `outbox` می نشد. اثبات تراکنشی

Classification: `IMPLEMENTED_WITH_GAP`.

Partner Packing List

Partner Logistics Controller:

- `list` و `get` از `partner.packing_lists`؛
- `PATCH` به محدودشده
- `audit log` برای `update`؛
- `merge metadata`.

تغییر مستقیم را فیلدها از بزرگی طی نیز `update` مدل دهد. می پاسخ `notAvailable` پیام و `503` با `POST create` ام ندارد. `version immutable` و `version` می

Operation	Status
List/Get	VERIFIED_IMPLEMENTED
Partner-scoped Patch/Audit	IMPLEMENTED_WITH_GAP
Create	PLANNED_NOT_IMPLEMENTED

Commercial Invoice

کند: می ذخیره را اطلاعات این واقعی Migration

- unique invoice number؛
- Packing List, Shipment, Quotation و Order references؛
- Partner و Customer؛
- invoice date, currency, amount؛
- Incoterm و payment terms؛
- ports, origin و destination؛
- HS code و declared value؛
- status, issuer, lineItems و metadata.

حال: با این کند. می اجرا را Partner-scoped patch و Controller list, get

- create با 503 می پاسخ
- generate-from-packing-list با 503 می پاسخ
- issue/sign/version/supersede workflow وجود ندارد.

Canonicalization موردنیاز

1. شود. انتخاب Packing List System of Record کی
2. Document identity, version, issuance, signature, supersession و cancellation تعریف شوند.
3. JSON line items و relational boxes reconciliation شوند.
4. quantities/weights با Production, Quality Release, PO و Shipment تطبیق داده شوند.
5. HS-code suggestion از legal classification باشد.
6. discrepancy/correction صورت به Aggregate ثبت شود، mutable patch نه.

03 — Export Customs and Legal Release

Provider implementation

کند: می ایجاد داده‌ها این با را partner.export_declarations واقعی Migration

- declaration number و customs office!
- shipment, commercial invoice و packing list!
- Partner و declarant!
- transport, vessel, voyage و container!
- ports و destination country!
- package/weight/value/currency!
- status, customs response و metadata.

کند. می ثبت audit و status history و دارد را submit و create, update, detail, list رفتارهای Partner Logistics Controller

این است. داده انجام submission action کاربر کرده ثبت سیستم است: platform fact یک فقط SUBMITTED ثبت است. کرده release را کالا یا پذیرفته را اظهارنامه Customs Authority کند: می اثبات وضعیت

دهند. می پاسخ 503 با هنوز Customs document add/remove

Tajerestan Customs

دارد: را زیر مسیرهای Tajerestan

- POST /api/v1/trade/customs/{poId}/clear
- get status!
- list و stats.

کند. می منتظر trade.customs.cleared و دهمی تغییری را Purchase Order روی customs فیلدهای مستقیم Service
نشده بررسی های فایل در declaration aggregate یا signed authority response External customs adapter.

است: نادرست رابطه این

Application user sets PO.customsStatus = CLEARED

Customs authority issued legal release

internal status, IMPLEMENTED_WITH_GAP؛ legal release, REFERENCE_ARCHITECTURE. ثبت برای بخش این

اصلی های Gap

- status! string آزاد هستند
- Partner ownership در همه SQL statement! شود. نمی دیده یک دست
- callback signature, response correlation, Outcome Unknown و revocation وجود ندارد.
- document attachment executable نیست.
- history در JSON mutable ذخیره شود.
- Partner declaration و Tajerestan PO customs مدل دو

canonical تصمیم

کند: جدا را Truth! این باید Export Customs

1. ExportDeclarationPrepared
2. ExportDeclarationSubmitted
3. AuthorityResponseObserved
4. ExportLegalReleaseObserved
5. AuthorityDecisionCorrectedOrRevoked

6. derived shipment-readiness perspective

است. معتبر authority evidence با فقط Legal release

04 — Import Customs, Duties, and Release

Evidence maturity

است. Wave 02 در شده بررسی های بخش تری ن بالغ از یکی Import Customs

کنند: می ایجاد را هاجدول این Migration V16

- partner.import_declarations
- partner.wms_inbound_orders
- partner.wms_inbound_items
- partner.warehouse_receipts

دارد: lifecycle Partner-scoped یکی Controller

- list و detail:
- create به متعلق Shipment از Partner:
- generic update در فقط DRAFT و CUSTOMS_HOLD:
- submit: اختصاصی!
- mark-duty-paid: اختصاصی!
- mark-cleared: اختصاصی!
- status history و audit:
- sync وضعیت Shipment به CUSTOMS_CLEARED.

Classification: IMPLEMENTED_WITH_GAP.

مثبت های کنترل

- شود. می گرفته JWT principal از Partner ID
- شوند. می محدود Partner ID با read/write از بسیری
- باشد. Partner به متعلق Shipment کند می بررسی create
- است. ممنوع PATCH generic از terminal status
- submit از فقط DRAFT
- دارد. نی از شده لیکن Shipment به clear
- شود. می ثبت history و audit snapshot

مهم های Gap

1. Migration destructive رفتار این کند: می recreate و DROP را partner.import_declarations عمداً Migration
2. کند. می غیرمنطقی را lifecycle که است مجاز نیز CLEARED وضعیت از mark-duty-paid
3. کند. نمی block را عملیات و کند می ایجاد warning فقط null یا صرف duty/tax
4. نشده. اثبات خارجی authority response و است application action یکی mark-cleared
5. کند. می تغییری direct cross-table mutation با Shipment status
6. ندارند. JDBC update optimistic version predicate
7. است. status history در JSON mutable
8. دارند. پوشانی هم Warehouse/Inventory ownership با Provider WMS tables

canonical مدل

باشند: مستقل باید زیر های Truth

- ImportDeclarationSubmitted — platform fact:
- DutyAssessmentObserved — authority observation:
- DutyPaymentIntent و DutyPaymentOutcome — Finance/Payment:
- ImportReleaseObserved — authority observation:
- ShipmentCustomsPerspectiveChanged — projection.

بسیار از legal clearance بتواند authority evidence بدون نباید برنامه کاربردی هیچ

05 — Transport Booking, Handover, Shipment, and Tracking

اجرای های Context

دارد: وجود اصلی Context دو

1. provider-service برای Partner logistics و Booking؛
2. shipping-service برای shipping.trade_shipments و API حمل B2B.

اند. نشده canonical هنوز Custody و Handover، Carrier Observation، Shipment، Booking مرزهای

Booking

Migration partner.bookings اطلاعات carrier, mode, vessel/flight, container, ports, dates, transit estimate, agent, confirmation, status, Partner و metadata را ذخیره می کند.

Export Handover Portal از API واقعی /api/v1/logistics/bookings استفاده می کند این و statel را نمایش می دهد:

- REQUESTED
- CONFIRMED
- LOADED
- HANDED_OVER
- CANCELLED

Portal را Commercial Invoice و Packing List های reference و seal number، loading time، contact، pickup address، همچنین journey می گیرد.

نیست. مغتبر Custody Transfer تنهائی به HANDED_OVER اما است، مفید اجرای journey این

Trade Shipment

TradeShipmentEntity شامل این داده ها است:

- Purchase Order و Trade Order reference؛
- carrier و tracking number؛
- transport mode و booking reference؛
- master/house bill؛
- container count و forwarder؛
- origin/destination؛
- status، ETA و actual delivery؛
- weight، volume و package count؛
- metadata و tenant.

Service رفتار create، list/filter، get، update، tracking append، stats و Kafka send را دارد. /api/shipping/trade/shipments را ارائه می دهد.

Classification: IMPLEMENTED_WITH_GAP.

کد بحرانی های یافته

- واقعی. tenant نه شود، می داده قرار Shipment UUID برابر tenantId، Shipment Item ساخت هنگام
- list/get/update/track در Service tenant-scoped شده بررسی
- items با findAll() و گرفته in-memory و
- status string بدون transition guard است.
- tracking history در metadata JSON mutable ذخیره می شود.
- دهد. نمی قرار payload داخل را آن اما گیر، می eventType پارامتر publish
- بماند. موفق Event بدون تواندمی DB transaction و log catch و Kafka خطای
- outbox نشد. پیدا تراکنشی

- delayed count صفر برابر hard-code شده است.
- observation های carrier فاقد source event ID, observed/received time, confidence و correction lineage هستند.

Custody

نشد. پیدا Custody مستقل Aggregate اما شود، می دیده Portal و Booking در HANDED_OVER وضعیت

بشود: موارد این شامل باید مع تبر Custody

- Cargo/Package Set؛ مشخص
- handing party و mandate؛
- receiving party و mandate؛
- time و location؛
- seal, package count و weight observation؛
- attestation یا طرف دو exception؛
- refusal/discrepancy؛
- correction؛
- onward transfer یا custody termination.

:نتیجه Booking و Shipment IMPLEMENTED_WITH_GAP؛ Custody REFERENCE_ARCHITECTURE.

06 — Warehouse Receiving and Receipt

Warehouse Service

کند: می ذخیره را اطلاعات این AsnEntity

- ASN number و reference!
- Purchase Order, warehouse, tenant و supplier!
- carrier و tracking!
- pallet count, line count و expected quantity!
- estimated/actual arrival!
- dock appointment!
- line items در JSON!
- status و timestamps.

دارد: را زیر رفتارهای AsnService

- create ASN!
- update status!
- link به Dock Appointment!
- get/list/pending!
- conflict detection برای dock!
- create/check-in/complete/cancel appointment!
- receiving summary.

دهد. می ارائه را /api/v1/warehouse/receiving مسیریهای AsnController

Classification: IMPLEMENTED_WITH_GAP.

Provider WMS

کند: می ایجاد را ها این نیز Provider schema در Migration V16

- wms_inbound_orders
- wms_inbound_items
- warehouse_receipts

است. CONFLICT_REQUIRES_CANONICALIZATION آن وضعیت و دارد پوشانی هم Warehouse Service با مدل این

قطعی های Gap

- ASN status می پذیرفته درخواست string از مستقیماً
- get/update/link بر اساس ID, tenant scope ندارند.
- optimistic lock روی ASN یی Dock Appointment
- Event/outbox پیدای Inventory به Receiving اتصال برای
- line items در TEXT/JSON می ذخیره
- summary dock door کل تعداد
- receipt line scan, package identity, damage evidence, quarantine, discrepancy و signed proof در Service شده بررسی
- دارند. وجود
- Provider warehouse receipt به تواندمی System of Record
- Delivery/POD و Warehouse Receipt از وضوح به هم

canonical مرز

مالک را Stock Position بای Inventory باشد. مالک را Receipt Observation و Physical Receiving Execution بای Warehouse

هدف: جریان

ASN + Dock Appointment
→ Arrival Observation
→ Package/Line Receiving
→ Damage/Discrepancy/Quarantine
→ Authorized Warehouse Receipt
→ Inventory Posting Command
→ Stock Position and Ledger

07 — Inventory Synchronization, Reservation, and QC Disposition

Stock Model

دارد: composite identity کی StockBalanceEntity

- tenant
- company
- warehouse
- SKU
- bin
- batch

شوند: می نگه‌داری جدا زیر مقادیر

- on hand
- available
- reserved
- damaged
- QC hold
- blocked
- in transit

کند. می فراهم Entity سطح در را @Version optimistic locking وجود

Inventory Posting

دارد: را زیر رفتاره‌ای transactional InventoryPostingService

- Receipt!
- Transfer!
- Adjustment!
- QC Hold! به Available انتقال
- QC PASS! به Available
- QC FAIL! یا Blocked یا Damaged به QC
- Stock Ledger ثبت

Reservation

ReservationService:

- quantity validate می را مثبت
- balance های tenant/company/warehouse/SKU خواند: می را
- کند: می انت‌خاب را کافی available دارای balance اولین
- کند: می زیاد را Reserved و کم را Available
- سازد: می state ACTIVE با Reservation
- گرداند. برمی Available به را آزادش‌دنی مقدار Release در

Classification: IMPLEMENTED_WITH_GAP.

قوت نقاط

- stock states صریح و تفکیکی
- tenant/company در composite key!
- optimistic locking!
- ledger دارای source و idempotency field!
- insufficient-stock check برای reservation و QC hold!

- Reservation Entity. مستقل.

بحرانی های Gap

1. نیستند. اعتماد قابل authority اطلاعات این گیرد؛ می query parameter از را user و company, tenant, Controller
2. است. GET /stock/ledger در Controller بدون شده بررسی tenant filter
3. Posting API های هم در عمومی هم Controller نشدند دیده caller/orchestration و نشدند
4. idempotency key در ledger می ذخیره شود، اما قبل از balance mutation duplicate lookup انجام نمی شود.
5. Transfer و Adjustment quantity توانند می
6. Reservation اولی balance مناسب را بدون policy مشخص
7. retry strategy برای optimistic-lock conflict پیدا نشد.
8. Receipt, good quantity را فوراً Available و گندمی quarantine/quality gate اجباری
9. Event/outbox برای Inventory facts پیدا نشد.
10. Provider WMS و Warehouse Receiving با Inventory ownership مرز روشن ندارند.

لازم اصلاح

Warehouse Receipt باید governed برای Inventory باشد. Availability با باید Policy مانده های Quality, Quarantine, Lot, Expiry, Compliance و Channel Eligibility receipt با صرفاً نه شود؛ محاسبه

08 — Cross-Domain Contradictions and Canonicalization Register

ID	Conflict	Required canonical decision
C-201	Quality به software gates, mock inspection و stock QC کندن می اشاره	Trade Quality Domain ایجاد مستقل شود
C-202	Packing List در Tajerestan و Partner وجود دارد	System of Record و migration direction شود تعینی
C-203	Commercial Invoice table/read/update ندارد create/generate ولی دارد،	Document lifecycle شود کامل
C-204	Customs declaration PO یا تغیری با action شود Cleared تواندمی	legal release فقط authority observation باشد
C-205	Import Controller مستقیماً Shipment table را تغیری را	Event/contract یا ownership مشترک شود تغیری رسمی
C-206	Booking و Trade Shipment بدون boundary اند شده جدا واضح	Booking, Shipment, Leg و Observation ownership شود تعینی
C-207	HANDED_OVER جای به Custody evidence شود می استفاده	CustodyRelationship ایجاد مستقل شود
C-208	Receiving Warehouse Service در هم و Provider WMS وجود دارد	Warehouse owner و Provider projection شود مشخص
C-209	Inventory Receipt Available بل اف اصله شود می	quarantine/quality/channel policy اعمال شود
C-210	tenant field ولی دارد وجود request parameter ی unscoped lookup دارند	trusted identity context و isolation test
C-211	Eventlo بدون outbox و فرستاده swallow شود می	outbox/inbox است ان دارد
C-212	status history در metadata JSON mutable است	append-only transition journal ایجاد شود

09 — Implementation Status Matrix

Area	Narrow verified behavior	Classification	Main blocker
Trade Quality Inspection	Mock inspection UI only	REFERENCE_ARCHITECTURE	Aggregate/API/Evidence missing
Software Quality Gates	Metrics, gates, reports	VERIFIED_IMPLEMENTED	Different Domain
Inventory QC Hold	Hold and disposition	PARTIAL_IMPLEMENTATION	Inspection Decision absent
Tajerestan Packing List	Entity, create/get/list/Event	IMPLEMENTED_WITH_GAP	identity, weights, versioning
Partner Packing List	list/get/patch/audit	IMPLEMENTED_WITH_GAP	create unavailable, duplicate owner
Commercial Invoice	table/list/get/patch	PARTIAL_IMPLEMENTATION	create/generate/issue missing
Export Declaration	table/create/list/update/submit	IMPLEMENTED_WITH_GAP	authority evidence/doc operations
PO Customs	mutable PO status and Event	IMPLEMENTED_WITH_GAP	not legal release
Import Declaration	full operational lifecycle	IMPLEMENTED_WITH_GAP	manual authority/state/migration gaps
Booking	table and real Portal/API	IMPLEMENTED_WITH_GAP	custody boundary
Trade Shipment	entity/API/tracking/Kafka	IMPLEMENTED_WITH_GAP	tenant and Event reliability
Custody Transfer	HANDED_OVER status only	CLIENT_CONTRACT_PRESENT	no bilateral evidence
Warehouse ASN/Dock	create/status/schedule/check-in	IMPLEMENTED_WITH_GAP	scoping, receipt evidence, Event
Provider WMS	tables and references	VERIFIED_IMPLEMENTED schema	competing owner/runtime verification
Inventory	balance/posting/QC/reservation	IMPLEMENTED_WITH_GAP	auth, idempotency, invariants

10 — Wave 02 Issue Register

ID	Severity	Area	Finding	Required action
W2-001	Critical	Quality	Inspection page uses static mock records	Replace with real API or label as demo.
W2-002	Critical	Quality	No canonical trade Inspection/Measurement/Release aggregate	Create Quality Domain and authority model.
W2-003	High	Quality	Warehouse client claims /quality/status side effects; backend target not located	Resolve route and add integration tests.
W2-004	Critical	Packing	Tajerestan and Partner packing lists compete	Declare System of Record and migration.
W2-005	High	Packing	PackingListService calculates no net weight	Extend request model and validate totals.
W2-006	Critical	Packing	Default UUID used as supplier and company	Use trusted tenant/organization context.
W2-007	High	Documents	Commercial Invoice create/generate endpoints return 503	Implement governed creation.
W2-008	High	Documents	Issued document version/signature model absent	Add immutable versions and signing.
W2-009	Critical	Customs	PO customs clear can be mistaken for legal release	Require authority evidence and rename internal action.
W2-010	High	Export Customs	Customs document endpoints return 503	Implement evidence attachment.
W2-011	High	Export Customs	Partner predicate not visible in all SQL operations	Enforce ownership on every query/update.
W2-012	Critical	Import Customs	Migration drops import_declarations	Add environment/data gate and non-destructive migration.
W2-013	High	Import Customs	CLEARED may transition to DUTY_PAID	Correct state machine.
W2-014	High	Import Customs	Zero duty/tax values only produce warnings	Require policy/exception approval.
W2-015	Critical	Import Customs	Manual mark-cleared lacks authority adapter evidence	Store signed authority observation.
W2-016	High	Shipping	Shipment item tenantId is assigned shipment UUID	Correct code and migrate data.
W2-017	Critical	Shipping	Shipment reads/writes are not visibly tenant-scoped	Apply trusted tenant context and isolation tests.
W2-018	High	Shipping	Kafka failure is swallowed after DB mutation	Use transactional outbox.
W2-019	High	Shipping	eventType is not included in payload	Use governed Event envelope.
W2-020	High	Shipping	Status is unrestricted string	Add typed transitions.

ID	Severity	Area	Finding	Required action
W2-021	High	Shipping	Tracking history is mutable JSON	Create qualified observation journal.
W2-022	Critical	Custody	HANDED_OVER status is not custody proof	Implement bilateral custody evidence.
W2-023	Critical	Warehouse	ASN ID operations lack visible tenant scoping	Scope repository methods.
W2-024	High	Warehouse	ASN status accepts arbitrary string	Use typed Commands and guards.
W2-025	Medium	Warehouse	Receiving summary assumes ten dock doors	Model facility capacity.
W2-026	Critical	Warehouse	Provider WMS and Warehouse Service overlap	Canonicalize ownership.
W2-027	Critical	Inventory	Tenant/company/user are accepted via query parameters	Derive from auth context.
W2-028	Critical	Inventory	Ledger endpoint appears globally unscoped	Add tenant/warehouse filtering and authorization.
W2-029	High	Inventory	Idempotency key is stored but not checked first	Add duplicate guard before mutation.
W2-030	High	Inventory	Transfer/adjustment can create negative stock	Add invariants and DB constraints.
W2-031	High	Inventory	Receipt immediately increases available stock	Apply quarantine/quality policy.
W2-032	Medium	Inventory	Reservation uses first matching balance	Define deterministic allocation policy.
W2-033	High	Cross-domain	No transactional outbox found for reviewed flows	Standardize outbox/inbox.
W2-034	Medium	Testing	Test-file existence is not execution evidence	Run CI and attach logs/results.

11 — Wave 02 Traceability Matrix

Evidence ID	Repository path	Blob SHA	Classification
QUALITY-UI-MOCK	apps/quality-dashboard/ app/inspections/page.tsx	ca8b5c61b956a6d96ea80101806074bdf06c093	VERIFIED - IMPLEMENTED
QUALITY-SOFTWARE-GATES	backend/domain/ engagement/dashboard- service/src/main/java/ com/allinb2c/engagement/ dashboard/quality/ controller/ QualityController.java	6410c6402611f0d78e82cff2e74801504dd47b1a2	VERIFIED - IMPLEMENTED
QUALITY-QC-WORKBENCH	apps/warehouse-operator/ components/qc-hold/qc- hold-workbench.tsx	fc4a0f3087df0a08f97b4ceec6246739d0953af5	UNVERIFIED
PACKING-ENTITY	backend/domain/commerce/ tajerestan-services/src/ main/java/com/allinb2c/ commerce/tajerestan/ entity/ PackingListEntity.java	5317c771765edadf8483d0a8827627d1d56	VERIFIED - IMPLEMENTED
PACKING-SERVICE	backend/domain/commerce/ tajerestan-services/src/ main/java/com/allinb2c/ commerce/tajerestan/ service/ PackingListService.java	7502b07ed6b29882c904ac698012667e745f6a	IMPLEMENTED WITH GAP
COMMERCIAL-INVOICE-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V7__phase_3c_ commercial_invoices.sql	9c8133be1f5805cb64af02177089c1fb_d08f25	VERIFIED - IMPLEMENTED
PARTNER-LOGISTICS	backend/domain/commerce/ provider-service/src/ main/java/com/allinb2c/ commerce/provider/api/ PartnerLogisticsCon- troller.java	8e49184334fb4b87228876c38414045915d00115	IMPLEMENTED WITH GAP
EXPORT-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V10__phase_4b_ export_declarations.sql	5f110ad3c3ad3d5cfb609c4aee7583f160210b74b	VERIFIED - IMPLEMENTED
TAJERESTAN-CUSTOMS	backend/domain/commerce/ tajerestan-services/src/ main/java/com/allinb2c/ commerce/tajerestan/ service/CustomsService. java	7fcb117de233e9900881479d418964b5c717d4027	IMPLEMENTED WITH GAP
IMPORT-CONTROLLER	backend/domain/commerce/ provider-service/src/ main/java/com/allinb2c/ commerce/provider/api/ PartnerImportController. java	40a0fb1f289fea319a78229459102b1b42e400857	IMPLEMENTED WITH GAP
IMPORT-WMS-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V16__phase_6a_ import_wms.sql	c32d37a25ad374acf6dc9bc3561401b405d1180	IMPLEMENTED WITH GAP

Evidence ID	Repository path	Blob SHA	Classification
TRADE-SHIPMENT-ENTITY	backend/domain/ fulfillment/shipping- service/src/main/java/ com/allinb2c/ fulfillment/shipping/ domain/ TradeShipmentEntity.java	dbd89d171bcf44274d46dd75b17f480641b1c1e	VERIFIED-IMPLEMENTED
TRADE-SHIPMENT-SERVICE	backend/domain/ fulfillment/shipping- service/src/main/java/ com/allinb2c/ fulfillment/shipping/ service/ TradeShipmentService. java	3e663f6f02c8ab0adaa6b125816b142085e5b97dH_GAP	IMPLEMENTED WITH_GAP
TRADE-SHIPMENT- CONTROLLER	backend/domain/ fulfillment/shipping- service/src/main/java/ com/allinb2c/ fulfillment/shipping/ controller/ TradeShipmentController. java	f4e2677a455a200240629367a481f59751b1e1	VERIFIED-IMPLEMENTED
BOOKING-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V9__phase_4b_ bookings.sql	cbce5a59c3be64d2f1d731cfd8461931ed43b3a68	VERIFIED-IMPLEMENTED
EXPORT-HANDOVER-PORTAL	apps/partner-portal/app/ partner/export-handover/ page.tsx	b4d1831a45b9cd93e0f2cd92ec1755b03407b04c5	VERIFIED-CONTROL-PRESENT
ASN-ENTITY	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ domain/AsnEntity.java	2cf9087a494877821f2488cae72ad9a861b1e1	VERIFIED-IMPLEMENTED
ASN-SERVICE	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ service/AsnService.java	6fcd4eb6f1e92262f73106f08a1b41e9681cd51aH_GAP	IMPLEMENTED WITH_GAP
ASN-CONTROLLER	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ controller/ AsnController.java	94aabea2d7743182a8de8d45476c57a164102e1	VERIFIED-IMPLEMENTED
STOCK-BALANCE	backend/domain/ fulfillment/inventory- service/src/main/java/ com/allinb2c/ fulfillment/inventory/ model/entity/ StockBalanceEntity.java	4b8cfad0e6cfec626c5c6f0c785d134ed714b68	VERIFIED-IMPLEMENTED
INVENTORY-POSTING	backend/domain/ fulfillment/inventory- service/src/main/java/ com/allinb2c/ fulfillment/inventory/ service/ InventoryPostingService. java	a7efd075874daee8e6a849cb61fb2a165b7b8967H_GAP	IMPLEMENTED WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
INVENTORY-CONTROLLER	backend/domain/ fulfillment/inventory- service/src/main/java/ com/allinb2c/ fulfillment/inventory/ controller/ InventoryController.java	02b1c4dce7367fe41e733fe5315a1726a7e03561	IMPLEMENTATION_WITH_GAP
RESERVATION-SERVICE	backend/domain/ fulfillment/inventory- service/src/main/java/ com/allinb2c/ fulfillment/inventory/ service/ ReservationService.java	fec768ef4cfd702328660bccab01f4e73268506	IMPLEMENTATION_WITH_GAP

12 — Remediation and Verification Plan

Priority 0 — False Authority از جلوگیری

1. شود. نامیده legal customs release نباید declaration application status یا PO
2. شود. پذیرفته authority عنوان به request از tenant/company/user
3. شوند. tenant-scoped کاملاً Partner SQL و Shipping, Warehouse, Inventory
4. شود. canonical Customs و Shipment Warehouse Receipt, Packing List، مالکیت
5. شود. migrate موجود داده و اصلاح Shipment Item اشتباه tenant assignment

Priority 1 — Truth Preservation و Execution Safety

1. شود. اضافه document version/signature immutable
2. شود. ایجاد Inventory و Warehouse, Shipping, Customs, Trade Documents برای outbox/inbox
3. شود. جایگزین journal append-only با JSON history
4. شود. بررسی stock mutation از قبل idempotency
5. شود. اضافه state transition guard و optimistic version predicate
6. باشد. correction و source reference، observed time، received time دارای external outcome

Priority 2 — Capability تکمیل

1. شود. ساخت Trade Quality Aggregate
2. شود. سازی پیاده PL-to-CI reconciliation و Commercial Invoice create
3. شود. ساخت ingestion authority-response و customs document
4. شود. پیاده attestations با CustodyRelationship
5. شود. متصل Inventory به typed contract با Warehouse Receipt
6. شود. اعمال availability از قبل channel eligibility و quarantine

Status ارتقای برای لازم Evidence

- migration empty و upgrade database:
- tenant-isolation tests:
- concurrent posting/reservation tests:
- duplicate/replay tests:
- outbox failure/recovery:
- customs/provider signature verification:
- cross-document reconciliation:
- Portal E2E بدون mock fallback:
- audit reconstruction و backup/restore.

13 — Repository Evidence Wave 03 Plan

کند: بررسی را Customer-facing Commerce م سری باید Wave 03

1. Product Catalog و Product Identity
2. Pricing و Price Version
3. Marketplace Offer و Publication Eligibility
4. Cart و Checkout
5. Customer Order Acceptance/Cancellation
6. Allocation و Fulfilment
7. Pick, Pack و Dispatch
8. Last-Mile Delivery و Proof
9. Returns, Claims, Refunds و Disputes
10. Settlement, Reporting, ESG و AI Evaluation

کند. نمی اثبات را Implementation تنه‌ای به UI contract یا README, route, plan, اثبات قانون

GTOS Batch 08 — Repository Evidence Wave 02 Combined Edition

00 — Scope, Method, and Repository Snapshot

Field	Value
Repository	chinair88-prog/allinb2c
Evidence snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Handbook branch	gtos/handbook-repository-grounded-expansion-v2
Wave	Repository Evidence Wave 02
Scope	Quality: documents: customs: transport: warehouse: inventory
Evidence standard	شده بررسی اجرای فایل با شده پشتی بانی و محدود ادعای

بررسی روش

فایل نام وجود شدند. خوانده مستقیماً SQL و JavaScript و Java، منتخب های فایل سپس و شد انجام کد و جوی جست ابتدا نشد. محسوب Implementation تنه ایی به route ی

اند: شده حفظ تمایزها این هافصل تمام در

1. frontend presentation برابر در backend execution؛
2. software quality برابر در trade-goods quality؛
3. platform status برابر در external legal authority؛
4. کامل؛ lifecycle برابر در جدول وجود
5. duplicate؛ از واقعی جلوگیری برابر در idempotency key ذخیره
6. اعتماد؛ قابل tenant enforcement برابر در tenant field وجود
7. شده؛ کنترل state machine برابر در status string
8. outbox. و شده تضمین delivery برابر در Event publisher فراخوانی

هام محدودیت

- است. نشده اثبات snapshot این Deployment
- نیست. CI یا Test موفق اجرای معنی به Test فایل وجود
- مستقل. اجرای نه اند، شده بررسی source روی از migration؛
- شوند. authoritative توان ندی authority evidence با فقط warehouse و inspector، customs، outcome؛
- است. integration test نیازمند migration با runtime datasource هماهنگی و دارند JDBC direct؛ controller از بخشی

01 — Quality Inspection, QC Hold, and Release

اجرای نتایج

م تفاوت مفهوم سه نشد. پیدا شده بررسی شواهد در تجاری کالای کیفی بازرسی برای canonical و مستقل Aggregate اند: شده مخلوط Quality وازه با

1. می محلی ثابت آرایه یک از را Inspection های داده apps/quality-dashboard/app/inspections/page.tsx صفحه 1. خواند.
2. می مدیریت را report و gate metric، افزار/پروژه، نرم کیفی در dashboard-service در QualityController
3. دارند. stock disposition و QC Hold محدودتر فرآیند یک Warehouse و Inventory

Capability	Status
Trade inspection planning and evidence	REFERENCE_ARCHITECTURE
Quality Dashboard inspection page	FRONTEND MOCK
Software-quality gates/reports	VERIFIED_IMPLEMENTED، اما Domain تفاوت
Stock QC hold and disposition	PARTIAL_IMPLEMENTATION
Final release certificate	REQUIRES_REPOSITORY_REVIEW

Inspection صفحه

نمایش را pending و fail و pass های نتیجه و outgoing و final، in-process، incoming، های نوع Quality Dashboard صفحه شوند. می اجرا آرایه همان روی فقط نیز آمار و وجو هست فیلتر، اند. شده تعریف INSPECTIONS ثابت در رکوردها اما ده، می شود. معرفی backend inspection اثبات عنوان به نباید صفحه این

Software Quality Controller

دارد: را زیر رفتارهای /api/quality مسیر

- dashboard و score بر اساس projectId:
- trend های metric:
- ingestion و aggregation metric:
- check و status و failed conditions برای quality gate:
- export و schedule report، تولدی.

این Production Lot، Sample، Measurement، Specification، Inspector، Laboratory، Defect و Release Certificate فاقد مدل این نیست. کالای کیفی Stage به متعلق بنابراین است؛

Inventory QC واقعی رفتار

کند: می سازی پیاده را زیر رفتارهای InventoryPostingService

- QC Hold به Available از quantity انتقال:
- available quantity بودن کافی کنترل:
- disposition نوع PASS و Available به بازگشت:
- Damaged به انتقال و FAIL_DAMAGED:
- Blocked به انتقال و FAIL_BLOCKED:
- QC_HOLD، QC_PASS، QC_FAIL_DAMAGED و QC_FAIL_BLOCKED های ledger event ثبت

specification و sampling plan چه کرده، بازرسی کسی چه کنند می ثابت اما کنند، می ثابت را stock disposition رفتار این است. کرده صادر را release ای authority چه و شده ثبت های measurement چه شده، استفاده

QC Hold Workbench

Warehouse UI:

- کند؛ می است فاده `inventoryApi.listQcHolds()` از
- کند؛ می فراخوانی `inspect/release/reject` ای action با را `PATCH /api/proxy/quality/status`
- دارد؛ `PENDING`, `INSPECTING`, `RELEASED` و `REJECTED` ای state
- شود. می `Stock Adjustment` باعث `Reject` و `Inventory Restock` باعث `Release` کند می ادعا

backend متناظر `/quality/status` در شواهد این پیدا Wave ای نیست؛ پایداری این `CLIENT_CONTRACT_UN-VERIFIED` است.

مفرد Subjects

اند: لازم Subjects این هوز کیفیت Stage تکمیل برای

- `InspectionPlan`
- `Inspection`
- `Sample`
- `Measurement`
- `SpecificationVersion`
- `Defect`
- `Nonconformance`
- `CorrectiveAction`
- `QualityDecision`
- `ReleaseCertificate`
- `ReleaseRevocation`
- `Inspector/Laboratory mandate` و `calibration evidence`

اصلاحی اقدامات

1. باشد. داشته جدا `ownership` و گذار نام `Trade Quality` از `Software Quality`
 2. شود. گذار عملیات `Demo` صریحاً یا متصل واقعی API به `Mock inspection UI`
 3. شود. ایجاد `Inventory` از مسئول `Quality Domain`
 4. شود. متصل `Inventory disposition` به شده تایی `Command` طریقی از `Quality Decision`
 5. شوند. ثبت `revocation` و `rationale`, `evidence`, `inspector identity`, `specification version`
 6. شود. `integration-test` و مسئولند پیدا، `/quality/status` به مربوط `proxy`
-

02 — Packing List and Commercial Invoice

Ownership

دارد: وجود Packing List مدل دو

- `Tajerestan` با `JPA` و جدول `tb_packing_list`؛
- `Provider/Partner` با `direct JDBC` و جدول `partner.packing_lists`.

شود. می نگه‌داری `partner.commercial_invoices` در نیز `Commercial Invoice`

است. `CONFLICT_REQUIRES_CANONICALIZATION` برابر `Packing List` وضعیت، `System of Record` تعین زمان تا

Tajerestan Packing List

فیلدهاست: این شامل واقعی Entity

- `PL` شماره؛
- `Purchase Order reference` و `PI`؛
- `supplier`؛
- `box count`؛
- `net/gross weight` و `volume`؛
- `packing method` و `marks`؛
- `status` و `company`؛
- `child boxes`.

واقعی: Service

- کند؛ می تولید `PL-...` شماره
- کند؛ می جمع را `volume` و `gross weight`، `package count`
- کند؛ می ذخیره را `box`
- دارد؛ `list` و `get`
- کند. می منتشر `trade.packing-list.created` نام با Event

Gapهای

- `DEFAULT_COMPANY_ID` هم و `supplier` برای هم
- `totalNetWeight` می صرف
- `netWeightKg` هر `box` صرف
- `immutable issued version`، `signature` و `supersession` نشده
- `status transition guard` و `optimistic locking` دیده نشده
- `outbox` نشده. اثبات تراکنشی

Classification: `IMPLEMENTED_WITH_GAP`.

Partner Packing List

Partner Logistics Controller:

- `list` و `get` از `partner.packing_lists`؛
- `PATCH` به محدود شده
- `audit log` برای `update`؛
- `merge metadata`.

تغییر مستقیم را فیلدها از بزرگی طی نیز `update` مدل دهد. می پاسخ `notAvailable` پیام و `503` با `POST create` اما ندارد. `version immutable` و `version` دهد

Operation	Status
List/Get	VERIFIED_IMPLEMENTED
Partner-scoped Patch/Audit	IMPLEMENTED_WITH_GAP
Create	PLANNED_NOT_IMPLEMENTED

Commercial Invoice

کند: می ذخیره را اطلاعات این واقعی Migration

- unique invoice number؛
- Packing List, Shipment, Quotation و Order references؛
- Partner و Customer؛
- invoice date, currency, amount؛
- Incoterm و payment terms؛
- ports, origin و destination؛
- HS code و declared value؛
- status, issuer, lineItems و metadata.

حال: با این کنند. می اجرا را Partner-scoped patch و Controller list, get

- create با 503 می پاسخ
- generate-from-packing-list با 503 می پاسخ
- issue/sign/version/supersede workflow وجود ندارد.

Canonicalization موردنیاز

1. شود. انتخاب Packing List System of Record کی
2. Document identity, version, issuance, signature, supersession و cancellation تعریف شوند.
3. JSON line items و relational boxes reconciliation شوند.
4. quantities/weights با Production, Quality Release, PO و Shipment تطبیق داده شوند.
5. HS-code suggestion از legal classification باشد.
6. discrepancy/correction صورت به Aggregate ثبت شود، mutable patch نه.

03 — Export Customs and Legal Release

Provider implementation

کند: می ایجاد داده‌های این با `partner.export_declarations` واقعی Migration

- declaration number و customs office!
- shipment, commercial invoice و packing list!
- Partner و declarant!
- transport, vessel, voyage و container!
- ports و destination country!
- package/weight/value/currency!
- status, customs response و metadata.

کند. می ثبت `audit` و `status history` و دارد را `submit` و `update`, `detail`, `list`, رفتارهای `Partner Logistics Controller`

این است. داده انجام `submission action` کاربر کرده ثبت سیستم است: `platform fact` یک فقط `SUBMITTED` ثبت است. کرده `release` را کالا یا پذیرفته را اظهارنامه `Customs Authority` کند می اثبات وضعیت

دهند. می پاسخ 503 با هنوز `Customs document add/remove`

Tajerestan Customs

دارد: را زیر مسیرهای `Tajerestan`

- `POST /api/v1/trade/customs/{poId}/clear`
- `get status!`
- `list و stats.`

کند. می منتظر `trade.customs.cleared` و دهمی تغییری را `Purchase Order` روی `customs` فیلدهای مستقیماً `Service` نشد. دیده شده بررسی های فایل در `declaration aggregate` یا `signed authority response` `External customs adapter`

است: نادرست رابطه این

`Application user sets PO.customsStatus = CLEARED`

`Customs authority issued legal release`

`internal status, IMPLEMENTED_WITH_GAP`؛ است: `legal release, REFERENCE_ARCHITECTURE` برای بخش این

اصلی های Gap

- `status! string` آزاد هستند.
- `Partner ownership` همه در `SQL statement! کدست` شود. نمی دیده یک
- `callback signature, response correlation, Outcome Unknown` و `revocation` وجود ندارد.
- `document attachment executable` نیست.
- `history` در `JSON mutable` ذخیره می شود.
- `Partner declaration` و `Tajerestan PO customs` دو مدل

canonical تصمیم

کند: جدا را `Truth! این` باید `Export Customs`

1. `ExportDeclarationPrepared`
2. `ExportDeclarationSubmitted`
3. `AuthorityResponseObserved`
4. `ExportLegalReleaseObserved`
5. `AuthorityDecisionCorrectedOrRevoked`

6. derived shipment-readiness perspective

است. معتمد بر authority evidence با فقط Legal release

04 — Import Customs, Duties, and Release

Evidence maturity

است. Wave 02 در شده بررسی های بخش تری نبالغ از یکی Import Customs

کنند: می ایجاد را هاجدول این Migration V16

- partner.import_declarations
- partner.wms_inbound_orders
- partner.wms_inbound_items
- partner.warehouse_receipts

دارد: lifecycle Partner-scoped یکی Controller

- list و detail:
- create به متعلق Shipment از Partner:
- generic update در فقط DRAFT و CUSTOMS_HOLD:
- submit: اختصاصی!
- mark-duty-paid: اختصاصی!
- mark-cleared: اختصاصی!
- status history و audit:
- sync وضعیت Shipment به CUSTOMS_CLEARED.

Classification: IMPLEMENTED_WITH_GAP.

مثبت های کنترل

- شود. می گرفته JWT principal از Partner ID
- شوند. می محدود Partner ID با read/write از بساری
- باشد. Partner به متعلق Shipment کندمی بررسی create
- است. ممنوع PATCH generic از terminal status
- submit از فقط DRAFT
- دارد. نیاز شده لیکن Shipment به clear
- شود. می ثبت history و audit snapshot

مهم های Gap

1. Migration destructive رفتار این کنند: می recreate و DROP را partner.import_declarations عمداً Migration
2. کنند. می غیرمنطقی را lifecycle که است مجاز نیز CLEARED وضعیت از mark-duty-paid
3. کنند. نمی block را عملیات و کندمی ایجاد warning فقط null یا صرف duty/tax
4. نشده. اثبات خارجی authority response و است application action یکی mark-cleared
5. کنند. می تغییری direct cross-table mutation با Shipment status
6. ندارند. JDBC update optimistic version predicate
7. است. status history در JSON mutable
8. دارند. پوشانی هم Warehouse/Inventory ownership با Provider WMS tables

canonical مدل

باشند: مستقل باید زیر های Truth

- ImportDeclarationSubmitted — platform fact:
- DutyAssessmentObserved — authority observation:
- DutyPaymentIntent و DutyPaymentOutcome — Finance/Payment:
- ImportReleaseObserved — authority observation:
- ShipmentCustomsPerspectiveChanged — projection.

بپردازد. legal clearance بتواند authority evidence بدون نیاز به برنامه کاربردی هیچ

05 — Transport Booking, Handover, Shipment, and Tracking

اجرای های Context

دارد: وجود اصلی Context دو

1. provider-service برای Partner logistics و Booking؛
2. shipping-service برای shipping.trade_shipments و API حمل B2B.

اند. نشده canonical هنوز Custody و Handover، Carrier Observation، Shipment، Booking مرزهای

Booking

Migration partner.bookings اطلاعات carrier, mode, vessel/flight, container, ports, dates, transit estimate, agent, confirmation, status, Partner و metadata را ذخیره می کنند

داده می نمایش را state را این و کند می است فاده /api/v1/logistics/bookings واقعی API از Export Handover Portal

- REQUESTED
- CONFIRMED
- LOADED
- HANDED_OVER
- CANCELLED

را Commercial Invoice و Packing List های reference و seal number، loading time، pickup address، همچنین Portal می گیرد.

نیست. مغت بر Custody Transfer تنه ای به HANDED_OVER اما است، مفید اجرای journey این

Trade Shipment

هست: داده این شامل TradeShipmentEntity

- Purchase Order و Trade Order reference؛
- carrier و tracking number؛
- transport mode و booking reference؛
- master/house bill؛
- container count و forwarder؛
- origin/destination؛
- status، ETA و actual delivery؛
- weight، volume و package count؛
- metadata و tenant.

Service رفتار create, list/filter, get, update, tracking append, stats و Kafka send را دارد. /api/shipping/trade/shipments را ارائه می دهد.

Classification: IMPLEMENTED_WITH_GAP.

کد بحرانی های یافته

- واقعی. tenant نه شود، می داده قرار Shipment UUID برابر tenantId، Shipment Item ساخت هنگام
- نیستند. tenant-scoped شده بررسی Service در list/get/update/track
- شوند. می فی لتر in-memory و گرفته با findAll()
- است. transition guard بدون status string
- شود. می ذخیره metadata JSON mutable در tracking history
- دهد. نمی قرار payload داخل را آن اما گیر، می eventType پارامتر publish
- بماند. موفق Event بدون تواندمی DB transaction و log catch و Kafka خطای
- نشد. پیدا تراکنشی outbox

- delayed count صفر برابر hard-code شده است.
- observation های carrier فاقد source event ID, observed/received time, confidence و correction lineage هستند.

Custody

نشد. پیدا Custody مستقل Aggregate اما شود، می دیده Portal و Booking در HANDED_OVER وضعیت

بشود: موارد این شامل باید مع تبر Custody

- Cargo/Package Set؛ مشخص
- handing party و mandate؛
- receiving party و mandate؛
- time و location؛
- seal, package count و weight observation؛
- attestation یا طرف دو exception؛
- refusal/discrepancy؛
- correction؛
- onward transfer یا custody termination.

:نتیجه Booking و Shipment IMPLEMENTED_WITH_GAP؛ Custody REFERENCE_ARCHITECTURE.

06 — Warehouse Receiving and Receipt

Warehouse Service

کند: می ذخیره را اطلاعات این AsnEntity

- ASN number و reference!
- Purchase Order, warehouse, tenant و supplier!
- carrier و tracking!
- pallet count, line count و expected quantity!
- estimated/actual arrival!
- dock appointment!
- line items در JSON!
- status و timestamps.

دارد: را زیر رفتارهای AsnService

- create ASN!
- update status!
- link به Dock Appointment!
- get/list/pending!
- conflict detection برای dock!
- create/check-in/complete/cancel appointment!
- receiving summary.

دهد. می ارائه را /api/v1/warehouse/receiving مسیریهای AsnController

Classification: IMPLEMENTED_WITH_GAP.

Provider WMS

کند: می ایجاد را ها این نیز Provider schema در Migration V16

- wms_inbound_orders
- wms_inbound_items
- warehouse_receipts

است. CONFLICT_REQUIRES_CANONICALIZATION آن وضعیت و دارد پوشانی هم Warehouse Service با مدل این

قطعی های Gap

- ASN status می پذیرفته درخواست string از مستقیماً
- get/update/link بر اساس ID, tenant scope ندارند.
- optimistic lock روی ASN یی Dock Appointment
- Event/outbox پیدای Inventory به Receiving اتصال برای
- line items در TEXT/JSON می ذخیره
- summary dock door کل تعداد
- receipt line scan, package identity, damage evidence, quarantine, discrepancy و signed proof در Service شده بررسی
- دارند. وجود
- Provider warehouse receipt به تواندمی System of Record
- Delivery/POD و Warehouse Receipt از وضوح به هم

canonical مرز

مالک را Stock Position بای Inventory باشد. مالک را Receipt Observation و Physical Receiving Execution بای Warehouse

هدف: جریان

- ASN + Dock Appointment
 - Arrival Observation
 - Package/Line Receiving
 - Damage/Discrepancy/Quarantine
 - Authorized Warehouse Receipt
 - Inventory Posting Command
 - Stock Position and Ledger
-

07 — Inventory Synchronization, Reservation, and QC Disposition

Stock Model

دارد: composite identity کی StockBalanceEntity

- tenant
- company
- warehouse
- SKU
- bin
- batch

شوند: می نگه‌داری جدا زیر مقداری

- on hand
- available
- reserved
- damaged
- QC hold
- blocked
- in transit

کند. می فراهم Entity سطح در را @Version optimistic locking وجود

Inventory Posting

دارد: را زیر رفتاره‌ای transactional InventoryPostingService

- Receipt!
- Transfer!
- Adjustment!
- QC Hold! به Available انتقال
- QC PASS! به Available
- QC FAIL! یا Blocked یا Damaged به QC
- Stock Ledger ثبت

Reservation

ReservationService:

- quantity validate می کند؛ را مثبت
- balance های tenant/company/warehouse/SKU خواند؛ می را
- کند؛ می انت‌خاب را کافی available دارای balance اولین
- کند؛ می زیاد را Reserved و کم را Available
- سازد؛ می state ACTIVE با Reservation
- گرداند. برمی Available به را آزادش‌دنی مقدار Release در

Classification: IMPLEMENTED_WITH_GAP.

قوت نقاط

- stock states صریح و تفکیکی
- tenant/company در composite key!
- optimistic locking!
- ledger دارای source و idempotency field!
- insufficient-stock check برای reservation و QC hold!

- Reservation Entity. مستقل.

بحرانی های Gap

1. نیستند. اعتماد قابل authority اطلاعات این گیری؛ می query parameter از را user و company, tenant, Controller
2. است. GET /stock/ledger در Controller بدون شده بررسی tenant filter
3. Posting API های هم در عمومی هم Controller نشدند دیه caller/orchestration و نشدند
4. idempotency key در ledger می ذخیره شود، اما قبل از balance mutation duplicate lookup انجام نمی شود.
5. Transfer و Adjustment quantity توانند می
6. Reservation اولی balance مناسبت policy بدون را مناسبت
7. retry strategy برای optimistic-lock conflict پیدا نشد.
8. Receipt, good quantity را فوراً Available و کند می quarantine/quality gate اجباری
9. Event/outbox برای Inventory facts پیدا نشد.
10. Provider WMS و Warehouse Receiving با Inventory ownership روشن مرز ندارند.

لازم اصلاح

Warehouse Receipt باید governed برای Inventory باشد. Availability باید با Policy های Quality, Quarantine, Lot, Expiry, Compliance و Channel Eligibility محاسبه نه شود؛ receipt با صرفاً نه

08 — Cross-Domain Contradictions and Canonicalization Register

ID	Conflict	Required canonical decision
C-201	Quality به software gates, mock inspection و stock QC	ایجاد مستقل Trade Quality Domain شود
C-202	Packing List در Tajerestan و Partner وجود دارد	System of Record و migration direction شود تعینی
C-203	Commercial Invoice table/read/update ندارد create/generate ولی دارد،	Document lifecycle شود کامل
C-204	Customs declaration PO یا تغیری با action شود Cleared تواندمی	legal release فقط authority observation باشد
C-205	Import Controller مستقیماً Shipment table تغیری را دهدمی	Event/contract یا ownership مشترک شود تغیری رسمی
C-206	Booking و Trade Shipment بدون boundary اندشده جدا واضح	Booking, Shipment, Leg و Observation ownership شود تعینی
C-207	HANDED_OVER جای به Custody evidence شودمی استفاده	CustodyRelationship ایجاد مستقل شود
C-208	Receiving Warehouse Service در هم و Provider WMS وجود دارد	Warehouse owner و Provider projection شود مشخص
C-209	Inventory Receipt Available بل افاصله شودمی	quarantine/quality/channel policy اعمال شود
C-210	tenant field ولی دارد وجود request parameter ی unscoped lookup دارند	trusted identity context و isolation test
C-211	Eventلا بدون outbox فرستاده swallow شودمی	outbox/inbox استان دارد شود
C-212	status history در metadata JSON mutable است	append-only transition journal ایجاد شود

09 — Implementation Status Matrix

Area	Narrow verified behavior	Classification	Main blocker
Trade Quality Inspection	Mock inspection UI only	REFERENCE_ARCHITECTURE	Aggregate/API/Evidence missing
Software Quality Gates	Metrics, gates, reports	VERIFIED_IMPLEMENTED	Different Domain
Inventory QC Hold	Hold and disposition	PARTIAL_IMPLEMENTATION	Inspection Decision absent
Tajerestan Packing List	Entity, create/get/list/Event	IMPLEMENTED_WITH_GAP	identity, weights, versioning
Partner Packing List	list/get/patch/audit	IMPLEMENTED_WITH_GAP	create unavailable, duplicate owner
Commercial Invoice	table/list/get/patch	PARTIAL_IMPLEMENTATION	create/generate/issue missing
Export Declaration	table/create/list/update/submit	IMPLEMENTED_WITH_GAP	authority evidence/doc operations
PO Customs	mutable PO status and Event	IMPLEMENTED_WITH_GAP	not legal release
Import Declaration	full operational lifecycle	IMPLEMENTED_WITH_GAP	manual authority/state/migration gaps
Booking	table and real Portal/API	IMPLEMENTED_WITH_GAP	custody boundary
Trade Shipment	entity/API/tracking/Kafka	IMPLEMENTED_WITH_GAP	tenant and Event reliability
Custody Transfer	HANDED_OVER status only	CLIENT_CONTRACT_PRESENT	no bilateral evidence
Warehouse ASN/Dock	create/status/schedule/check-in	IMPLEMENTED_WITH_GAP	scoping, receipt evidence, Event
Provider WMS	tables and references	VERIFIED_IMPLEMENTED schema	competing owner/runtime verification
Inventory	balance/posting/QC/reservation	IMPLEMENTED_WITH_GAP	auth, idempotency, invariants

10 — Wave 02 Issue Register

ID	Severity	Area	Finding	Required action
W2-001	Critical	Quality	Inspection page uses static mock records	Replace with real API or label as demo.
W2-002	Critical	Quality	No canonical trade Inspection/Measurement/Release aggregate	Create Quality Domain and authority model.
W2-003	High	Quality	Warehouse client claims /quality/status side effects; backend target not located	Resolve route and add integration tests.
W2-004	Critical	Packing	Tajerestan and Partner packing lists compete	Declare System of Record and migration.
W2-005	High	Packing	PackingListService calculates no net weight	Extend request model and validate totals.
W2-006	Critical	Packing	Default UUID used as supplier and company	Use trusted tenant/organization context.
W2-007	High	Documents	Commercial Invoice create/generate endpoints return 503	Implement governed creation.
W2-008	High	Documents	Issued document version/signature model absent	Add immutable versions and signing.
W2-009	Critical	Customs	PO customs clear can be mistaken for legal release	Require authority evidence and rename internal action.
W2-010	High	Export Customs	Customs document endpoints return 503	Implement evidence attachment.
W2-011	High	Export Customs	Partner predicate not visible in all SQL operations	Enforce ownership on every query/update.
W2-012	Critical	Import Customs	Migration drops import_declarations	Add environment/data gate and non-destructive migration.
W2-013	High	Import Customs	CLEARED may transition to DUTY_PAID	Correct state machine.
W2-014	High	Import Customs	Zero duty/tax values only produce warnings	Require policy/exception approval.
W2-015	Critical	Import Customs	Manual mark-cleared lacks authority adapter evidence	Store signed authority observation.
W2-016	High	Shipping	Shipment item tenantId is assigned shipment UUID	Correct code and migrate data.
W2-017	Critical	Shipping	Shipment reads/writes are not visibly tenant-scoped	Apply trusted tenant context and isolation tests.
W2-018	High	Shipping	Kafka failure is swallowed after DB mutation	Use transactional outbox.
W2-019	High	Shipping	eventType is not included in payload	Use governed Event envelope.
W2-020	High	Shipping	Status is unrestricted string	Add typed transitions.

ID	Severity	Area	Finding	Required action
W2-021	High	Shipping	Tracking history is mutable JSON	Create qualified observation journal.
W2-022	Critical	Custody	HANDED_OVER status is not custody proof	Implement bilateral custody evidence.
W2-023	Critical	Warehouse	ASN ID operations lack visible tenant scoping	Scope repository methods.
W2-024	High	Warehouse	ASN status accepts arbitrary string	Use typed Commands and guards.
W2-025	Medium	Warehouse	Receiving summary assumes ten dock doors	Model facility capacity.
W2-026	Critical	Warehouse	Provider WMS and Warehouse Service overlap	Canonicalize ownership.
W2-027	Critical	Inventory	Tenant/company/user are accepted via query parameters	Derive from auth context.
W2-028	Critical	Inventory	Ledger endpoint appears globally unscoped	Add tenant/warehouse filtering and authorization.
W2-029	High	Inventory	Idempotency key is stored but not checked first	Add duplicate guard before mutation.
W2-030	High	Inventory	Transfer/adjustment can create negative stock	Add invariants and DB constraints.
W2-031	High	Inventory	Receipt immediately increases available stock	Apply quarantine/quality policy.
W2-032	Medium	Inventory	Reservation uses first matching balance	Define deterministic allocation policy.
W2-033	High	Cross-domain	No transactional outbox found for reviewed flows	Standardize outbox/inbox.
W2-034	Medium	Testing	Test-file existence is not execution evidence	Run CI and attach logs/results.

11 — Wave 02 Traceability Matrix

Evidence ID	Repository path	Blob SHA	Classification
QUALITY-UI-MOCK	apps/quality-dashboard/ app/inspections/page.tsx	ca8b5c61b956a6d96ea80101806074bdf06c093	VERIFIED - IMPLEMENTED
QUALITY-SOFTWARE-GATES	backend/domain/ engagement/dashboard- service/src/main/java/ com/allinb2c/engagement/ dashboard/quality/ controller/ QualityController.java	6410c6402611f0d78e82cff2e74801504dd47b1a2	VERIFIED - IMPLEMENTED
QUALITY-QC-WORKBENCH	apps/warehouse-operator/ components/qc-hold/qc- hold-workbench.tsx	fc4a0f3087df0a08f97b4ceec6246739d0953af5	UNVERIFIED - IMPLEMENTED
PACKING-ENTITY	backend/domain/commerce/ tajerestan-services/src/ main/java/com/allinb2c/ commerce/tajerestan/ entity/ PackingListEntity.java	5317c771765edadf8483d0a8827621d1d5611412	VERIFIED - IMPLEMENTED
PACKING-SERVICE	backend/domain/commerce/ tajerestan-services/src/ main/java/com/allinb2c/ commerce/tajerestan/ service/ PackingListService.java	7502b07ed6b29882c904ac698012667e745f6a	UNVERIFIED - WITH_GAP
COMMERCIAL-INVOICE-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V7__phase_3c_ commercial_invoices.sql	9c8133be1f5805cb64af02177089c1fb1d9f25	VERIFIED - IMPLEMENTED
PARTNER-LOGISTICS	backend/domain/commerce/ provider-service/src/ main/java/com/allinb2c/ commerce/provider/api/ PartnerLogisticsCon- troller.java	8e49184334fb4b87228876c38414045915d00115	UNVERIFIED - WITH_GAP
EXPORT-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V10__phase_4b_ export_declarations.sql	5f110ad3c3ad3d5cfb609c4aee7583f160210b743	VERIFIED - IMPLEMENTED
TAJERESTAN-CUSTOMS	backend/domain/commerce/ tajerestan-services/src/ main/java/com/allinb2c/ commerce/tajerestan/ service/CustomsService. java	7fcb117de233e9900881479d418964b5c71ed4027	UNVERIFIED - WITH_GAP
IMPORT-CONTROLLER	backend/domain/commerce/ provider-service/src/ main/java/com/allinb2c/ commerce/provider/api/ PartnerImportController. java	40a0fb1f289fea319a78229459102b1b42e400857	UNVERIFIED - WITH_GAP
IMPORT-WMS-DDL	backend/domain/commerce/ provider-service/src/ main/resources/db/ migration/V16__phase_6a_ import_wms.sql	c32d37a25ad374acf6dc9bc35614d1b405d1180	UNVERIFIED - WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
TRADE-SHIPMENT-ENTITY	backend/domain/fulfillment/shipping-service/src/main/java/com/allinb2c/fulfillment/shipping/domain/TradeShipmentEntity.java	dbd89d171bcf44274d46dd75b17f480641b1c1e	VERIFIED-IMPLEMENTED
TRADE-SHIPMENT-SERVICE	backend/domain/fulfillment/shipping-service/src/main/java/com/allinb2c/fulfillment/shipping/service/TradeShipmentService.java	3e663f6f02c8ab0adaa6b125816b142085e5b97d	IMPLEMENTED WITH_GAP
TRADE-SHIPMENT-CONTROLLER	backend/domain/fulfillment/shipping-service/src/main/java/com/allinb2c/fulfillment/shipping/controller/TradeShipmentController.java	f4e2677a455a200240629367a481f59751b1e1	VERIFIED-IMPLEMENTED
BOOKING-DDL	backend/domain/commerce/provider-service/src/main/resources/db/migration/V9__phase_4b_bookings.sql	cbce5a59c3be64d2f1d731cfd8461931ed4303a68	VERIFIED-IMPLEMENTED
EXPORT-HANDOVER-PORTAL	apps/partner-portal/app/partner/export-handover/page.tsx	b4d1831a45b9cd93e0f2cd92ec755b03c07b04c5	IMPLEMENTED-CONTROL-PRESENT
ASN-ENTITY	backend/domain/fulfillment/warehouse-service/src/main/java/com/allinb2c/fulfillment/warehouse/domain/AsnEntity.java	2cf9087a494877821f2488cae72ad9a861b1e1	VERIFIED-IMPLEMENTED
ASN-SERVICE	backend/domain/fulfillment/warehouse-service/src/main/java/com/allinb2c/fulfillment/warehouse/service/AsnService.java	6fcd4eb6f1e92262f73106f08a1b4f9681cd51a	IMPLEMENTED WITH_GAP
ASN-CONTROLLER	backend/domain/fulfillment/warehouse-service/src/main/java/com/allinb2c/fulfillment/warehouse/controller/AsnController.java	94aabea2d7743182a8de8d45476037a164102e1	VERIFIED-IMPLEMENTED
STOCK-BALANCE	backend/domain/fulfillment/inventory-service/src/main/java/com/allinb2c/fulfillment/inventory/model/entity/StockBalanceEntity.java	4b8cfad0e6cfec626c5c6f0c785d134ed714b68	VERIFIED-IMPLEMENTED
INVENTORY-POSTING	backend/domain/fulfillment/inventory-service/src/main/java/com/allinb2c/fulfillment/inventory/service/InventoryPostingService.java	a7efd075874daee8e6a849cb61f02a665b7b8967	IMPLEMENTED WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
INVENTORY-CONTROLLER	backend/domain/ fulfillment/inventory- service/src/main/java/ com/allinb2c/ fulfillment/inventory/ controller/ InventoryController.java	02b1c4dce7367fe41e733fe5315a126a0e03561	IMPLEMENTATION_WITH_GAP
RESERVATION-SERVICE	backend/domain/ fulfillment/inventory- service/src/main/java/ com/allinb2c/ fulfillment/inventory/ service/ ReservationService.java	fec768ef4cfd702328660bcca00f1e4e73268506	IMPLEMENTATION_WITH_GAP

12 — Remediation and Verification Plan

Priority 0 — False Authority از جلوگیری

1. شود. نامیده legal customs release نباید declaration application status یا PO
2. شود. پذیرفته authority عنوان به request از tenant/company/user
3. شوند. tenant-scoped کاملاً Partner SQL و Shipping, Warehouse, Inventory
4. شود. canonical Customs و Shipment Warehouse Receipt, Packing List، مالکیت
5. شود. migrate موجود داده و اصلاح Shipment Item اشتباه tenant assignment

Priority 1 — Truth Preservation و Execution Safety

1. شود. اضافه document version/signature immutable
2. شود. ایجاد Inventory و Warehouse, Shipping, Customs, Trade Documents برای outbox/inbox
3. شود. جایگزین journal append-only با JSON history
4. شود. بررسی stock mutation از قبل idempotency
5. شود. اضافه state transition guard و optimistic version predicate
6. باشد. correction و source reference، observed time، received time دارای external outcome

Priority 2 — Capability تکمیل

1. شود. ساخت Trade Quality Aggregate
2. شود. سازی پیاده PL-to-CI reconciliation و Commercial Invoice create
3. شود. ساخت authority-response ingestion و customs document
4. شود. پیاده attestations با CustodyRelationship
5. شود. متصل Inventory به typed contract با Warehouse Receipt
6. شود. اعمال availability از قبل channel eligibility و quarantine

Status ارتقای برای لازم Evidence

- migration empty و upgrade database:
 - tenant-isolation tests:
 - concurrent posting/reservation tests:
 - duplicate/replay tests:
 - outbox failure/recovery:
 - customs/provider signature verification:
 - cross-document reconciliation:
 - Portal E2E بدون mock fallback:
 - audit reconstruction و backup/restore.
-

13 — Repository Evidence Wave 03 Plan

کند: بررسی را Customer-facing Commerce م سری باید Wave 03

1. Product Catalog و Product Identity
2. Pricing و Price Version
3. Marketplace Offer و Publication Eligibility
4. Cart و Checkout
5. Customer Order Acceptance/Cancellation
6. Allocation و Fulfilment
7. Pick, Pack و Dispatch
8. Last-Mile Delivery و Proof
9. Returns, Claims, Refunds و Disputes
10. Settlement, Reporting, ESG و AI Evaluation

کند. نمی اثبات را Implementation تنه ای به UI contract ی README, plan, route, اس: ثابت قانون

Implementation and Evidence Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	را محدود ادعای و شده بررسی migration یا اجرای فایل کندمی اثبات
IMPLEMENTED_WITH_GAP	دارد مهم Gap ولی دارد وجود اجرای رفتار
PARTIAL_IMPLEMENTATION	شده اجرا Capability از بخشی فقط
CLIENT_CONTRACT_PRESENT	نباشد کامل است ممکن backend دارد؛ وجود Client contract
CLIENT_CONTRACT_UNVERIFIED	پیدا backend ولی کندمی ادعا endpoint یا Client side effect نشده
FRONTEND MOCK	کندمی استفاده محلی ثابت/ساختگی داده از UI
DOCUMENTATION_ONLY_CLAIM	دارد وجود plan یا سند فقط
CONFLICT_REQUIRES_CANONICALIZATION	دارند وجود رقیب های ownership یا Implementation
REFERENCE_ARCHITECTURE	Implementation ادعای بدون هدف؛ طراحی
REQUIRES_REPOSITORY_REVIEW	است لازم بیشتتری شواهد
TEST_FILE_DISCOVERED	نشده اثبات اجرا/نتیجه ولی شده پیدا Test فایل

GTOS Handbook — Batch 08

Repository Evidence Wave 02

کند:می تبدی code-grounded های گزارش به معماری عمومی متن از را زیر های حوزه بسته این

- Quality Inspection و QC disposition
- Packing List و Commercial Invoice
- Export Customs
- Import Customs و Duties
- Booking, Handover, Shipment و Tracking
- Warehouse Receiving
- Inventory Synchronization و Reservation

Field	Value
Repository	chinair88-prog/allinb2c
Snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Handbook Branch	gtos/handbook-repository-grounded-expansion-v2
Publication maturity	کتاب نه‌ای نسخه نه Review artifact!

و شده محدود اثبات قابل رفتار تری‌ن‌کوچک به فقط ادعا هر نیست. افزار نرم تکمیل معنی‌به Wave این نتی‌جه اند.نشده جعل خارجی حقیقت عنوان‌به تحویل و بازرسی حمل، گمرک، های outcome

Batch 08 Release Report

Field	Value
Release	Repository Evidence Wave 02
Repository snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Publication maturity	Code-grounded review artifact
Final handbook status	Not complete

Principal conclusions

- است. Trade Quality Inspection و stock disposition mock UI حد در
- دارد. وجود رقیب Packing List implementation دو
- نیست. کامل create/generate ولی دارد، Commercial Invoice persistence/read/update
- نیست. legal authority معادل application status ولی دارد، Export/Import workflow اجرای
- دارد. بحران Event delivery Gap و tenant isolation اما شده اجرا Shipping
- دارند. پوششانی هم Provider WMS و Warehouse ASN/Dock
- دارد. invariant Gap و request-derived identity، idempotency اما دارد، قوی مدل Inventory

Implementation and Evidence Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	را محدود ادعای و شده بررسی migration یا اجرای فایل کندمی اثبات
IMPLEMENTED_WITH_GAP	دارد مهم Gap ولی دارد وجود اجرای رفتار
PARTIAL_IMPLEMENTATION	شده اجرا Capability از بخشی فقط
CLIENT_CONTRACT_PRESENT	نباشد کامل است ممکن backend دارد؛ وجود Client contract
CLIENT_CONTRACT_UNVERIFIED	پیدا backend ولی کندمی ادعا endpoint یا Client side effect نشده
FRONTEND MOCK	کندمی استفاده محلی ثابت/ساختگی داده از UI
DOCUMENTATION_ONLY_CLAIM	دارد وجود plan یا سند فقط
CONFLICT_REQUIRES_CANONICALIZATION	دارند وجود رقیب های ownership یا Implementation
REFERENCE_ARCHITECTURE	Implementation ادعای بدون هدف؛ طراحی
REQUIRES_REPOSITORY_REVIEW	است لازم بیشتتری شواهد
TEST_FILE_DISCOVERED	نشده اثبات اجرا/نتیجه ولی شده پیدا Test فایل

Integrity Report

- Generated at: 2026-07-23T15:17:15.775567+00:00
- Markdown files before report: 18
- CSV files: 3
- Total words before report: 8,862
- Empty Markdown files: 0
- Source registry entries: 23
- Issue register entries: 34

Claim-discipline checks

```
{  
  "quality_mock_not_upgraded": true,  
  "customs_authority_boundary": true,  
  "shipping_tenant_bug_recorded": true,  
  "inventory_request_identity_gap": true,  
  "ownership_conflict_recorded": true,  
  "issue_register_present": true,  
  "source_registry_present": true  
}
```

Empty files

- None

Result

PASS

کند. نمی اثبات را افزار نرم production approval یا تکمیل و است بسته داخلی سازگاری و سلامت معنیه فقط PASS

GTOS Handbook — Batch 08

Repository Evidence Wave 02

کند:می تبدی code-grounded های گزارش به معماری عمومی متن از را زیر های حوزه بسته این

- Quality Inspection و QC disposition
- Packing List و Commercial Invoice
- Export Customs
- Import Customs و Duties
- Booking, Handover, Shipment و Tracking
- Warehouse Receiving
- Inventory Synchronization و Reservation

Field	Value
Repository	chinair88-prog/allinb2c
Snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Handbook Branch	gtos/handbook-repository-grounded-expansion-v2
Publication maturity	کتاب نه‌ای نسخه نه Review artifact!

و شده محدود اثبات قابل رفتار تری‌ن‌کوچک به فقط ادعا هر نیست. افزارنرم تکمیل معنی‌به Wave این نتی‌جه اند.نشده جعل خارجی حقیقت عنوان‌به تحویل و بازرسی حمل، گم‌رک، های outcome

Batch 08 Release Report

Field	Value
Release	Repository Evidence Wave 02
Repository snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Publication maturity	Code-grounded review artifact
Final handbook status	Not complete

Principal conclusions

- است. Trade Quality Inspection و stock disposition mock UI حد در
- دارد. وجود رقیب Packing List implementation دو
- نیست. کامل create/generate ولی دارد، Commercial Invoice persistence/read/update
- نیست. legal authority معادل application status ولی دارد، Export/Import workflow اجرای
- دارد. بحرانی Event delivery Gap و tenant isolation اما شده اجرا Shipping
- دارند. پوششانی هم Provider WMS و Warehouse ASN/Dock
- دارد. invariant Gap و request-derived identity، idempotency اما دارد، قوی مدل Inventory

00 — Scope, Method, and Snapshot

Field	Value
Repository	chinair88-prog/allinb2c
Snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Branch for handbook artifacts	gtos/handbook-repository-grounded-expansion-v2
Review date	2026-07-23
Scope	Catalog → Pricing → Checkout → Order → Fulfilment → Returns → Settlement → ESG → AI Evaluation

Evidence Method

ترین کوچک به فقط Classification شدند. خواننده مستقیماً Repository search از پس SQL و TypeScript و Java های فایل یافت. اختصاص دفاع قابل ادعای

اجباری: تمامی‌های

1. Product master در برابر Marketplace catalog entry و Offer؛
2. calculated price در برابر published/accepted Price Version؛
3. cart item price در برابر authoritative checkout quote؛
4. Order state در برابر Inventory/Payment/Fulfilment outcome؛
5. shipped/delivered status در برابر carrier observation یا Proof of Delivery؛
6. return inspection record در برابر inventory restock execution؛
7. settlement calculation در برابر payout execution؛
8. sustainability target در برابر independently verified ESG metric؛
9. evaluation submission در برابر trustworthy model evaluation and governance decision.

Limitations

- Runtime configuration, migration execution و CI results اجرا مستقیماً
- نیست. بودن pass معنی به Test file وجود
- External payment, carrier, customer delivery, sustainability assurance و model-evaluation authority اثبات نشده
- Header می محسوب trusted identity context, authority به اتصال صورت در فقط request field

01 — Product Catalog, Marketplace Catalog, and Offer

Three overlapping product representations

دارد: وجود اجرایی representation سه

1. catalog.products در product-catalog-service با ProductEntity.
2. marketplace-service با CatalogProductEntity در جدول همان روی دیگر read mapping یک.
3. marketplace.product_catalog_entries با tenant/company, unit price, MOQ, lead time و verification fields.

با: است برابر Product Master و Marketplace Offer برای وضعیت این

CONFLICT_REQUIRES_CANONICALIZATION

Product Catalog Service

کنند: می ارائه را زیر رفتارهای ProductService و ProductController

- list/search/filter؛
- get by ID و slug؛
- create/update/archive؛
- update status؛
- resell from purchase؛
- similar products؛
- tiered pricing.

ProductEntity شامل multilingual names, SKU, category, supplier, brand, origin, HS code, status, visibility, base/compare/cost price, dimensions, MOQ, tax, approval fields و source tracking است.

Classification: IMPLEMENTED_WITH_GAP.

Critical catalog findings

- ProductEntity tenant/company field ندارد.
- @Version روی Product دیده نشد.
- Controller authorization annotation و actor fallback برابر anonymous است.
- updateProduct و dedicated status workflow و کند تنظیم request از مستقیماً را status تواندمی
- updateProductStatus transition matrix یا role guard ندارد.
- resellFromPurchase ownership، Inventory possession، Quality eligibility یا legal resale eligibility را verify کند.
- Product Pricing و Marketplace Pricing که در حالی دارد، می نگه base/cost/compare price خود
- Product tier prices hard-coded هستند و Price Version مستقل نیستند.

Marketplace Product Catalog

marketplace.product_catalog_entries tenant/company-aware است و API CRUD, overview و category listing دارد. حال: بالاین

- create tenant/company می request از را پذیرد؛
- get/update/delete by ID tenant predicate آشکار
- isVerified و isFeatured از request قابل
- hard delete وجود دارد؛
- unitPrice داخل Catalog Entry ذخیره می شود.

Classification: IMPLEMENTED_WITH_GAP.

Missing Offer aggregate

یک Marketplace Offer canonical که Seller, Product Version, Price Version, Inventory Eligibility, Channel, Region, Terms, Availability Window و Publication Decision را immutable snapshot نهد، پیدا شده بررسی شود در کند، نشود.

Target boundary:

ProductMaster

→ ProductVersion

→ SellerAssortmentEligibility

→ PriceVersion

→ MarketplaceOffer

→ OfferPublicationDecision

→ CheckoutOfferSnapshot

02 — Pricing, Price Version, and Commercial Snapshot

Implemented pricing capabilities

دارد: وجود اصلی implementation دو

Pricing Service Target Price

TargetPriceController و TargetPriceService:

- create/update/delete policy!
- list by tenant/product!
- lookup active target price.

Marketplace Smart Pricing

SmartPricingService و SmartPricingController:

- Price Rule CRUD!
- Landed Cost CRUD!
- Market Price Snapshot!
- competitor averages!
- price calculation!
- pricing overview.

هستند. Strategies شامل Cost Plus, Market Based, Dynamic, Penetration, Skimming, Volume Tiered, Promotional و Bundle

Classification: IMPLEMENTED_WITH_GAP.

Critical pricing findings

- tenant/company در Target Price و Smart Pricing از request/query گرفته می شوند.
- get/update/delete rule by ID tenant-scoped نیستند.
- Target Price status string آزاد است.
- Catalog Product و Marketplace Catalog Entry قیمت نیز
- Category و Warehouse rule matching در engine با و true عمل می کنند.
- rule نتایج در شود! می اجرا simplified comment با و true عمل می کنند.
- scope از خارج تواندمی شود.
- Volume tiers hard-coded هستند.
- manual discount/surcharge از request پذیرفته می شود.
- response currency در Smart Pricing به USD hard-code شده است.
- competitor source authenticity, timestamp policy و observation confidence محدود است.
- Price Version, approval, publication, effective interval, supersession و accepted checkout quote پیداست.

Canonical pricing model

```
CostObservation
+ LandedCostVersion
+ MarketPriceObservation
+ PricingRuleVersion
→ PriceCalculation
→ PriceApprovalDecision
→ PriceVersion
→ OfferPrice
→ CheckoutPriceSnapshot
```

حق قیمت را Client unitPrice این که نه کنند، ذخیره rule/version/effective time با را Price Snapshot بای Order و Cart کنند. تلقی

03 — Cart, Checkout, and AI Insights

Cart implementation

دارند: را زیر رفتارهای CartService و CartController

- create/get cart by user or session!
- get by cart ID!
- add/update/remove item!
- coupon attach/remove!
- guest-to-user merge!
- clear and expiry cleanup.

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Cart security and commercial integrity gaps

- شوند. می پذیرفته request از cartId و sessionId، userId،
- ندارند. آشکار ownership check by cart ID get/update/delete
- پذیرد. می Client از را attributes و providerId، currency، unitPrice، name، SKU، فیلهای Cart item
- نشد. دیده از قبل add از Catalog/Offer/Inventory lookup
- شوند. نمی availability validate و channel eligibility، maximum quantity، MOQ
- انجام CouponService توسط بَعْدًا واقعی validation گویدمی صریحًا comment شود؛ می cart attach به فقط Coupon
- شود. می
- شود. می انجام Caller ارسالی sessionId/sessionId یا merge guest

Checkout implementation

CheckoutController:

- checkout start!
- cart-to-order conversion!
- checkout validation!
- order verification!
- payment confirmation!
- AI insights.

نمی انجام Catalog/Offer با واقعی comparison کنند. می منتقل Order Item به مستقیماً را Cart های قیمت checkout اما کنند. override را آن Order layer در TenantContext مگر شود، می Order وارد Checkout request از نیز tenantId شود.

confirm-payment، signed provider callback، idempotency، amount/currency reconciliation، Payment Intent ownership یا Order مستقیماً confirm-payment

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Simulated AI

کنند. می مغرفی AI decisions reference implementation that simulates AI analysis را صریحًا کد

Fraud score، ETA، payment suggestion، warehouse routing و upsell با ruleهای و hard-coded می ساخته شوند؛ برای شوند. می انتخاب فرضی های carrier و wh-special و wh-main های warehouse نمونه

Classification:

- AI checkout advice: SIMULATED_REFERENCE_IMPLEMENTATION
- persisted AI decision: PLANNED_NOT_IMPLEMENTED؛ کند. می log را آن فقط کد

Required checkout contract

Checkout: باید:

1. actor/cart ownership را verify کنند؛
2. Offer و Price Version را resolve دوباره کنند؛
3. availability و reservation را اجرا کنند؛
4. shipping/tax/discount را authoritative محاسبه کنند؛
5. quote expiry و idempotency داشته باشند؛
6. Payment Intent بسازد؛
7. Order Snapshot immutable ایجاد کنند؛
8. AI output را advisory و traceable نگه دارد.

04 — Customer Order and Workflow

Implemented order model

OrderEntity دارای:

- Order, Payment و Fulfilment status؛
- tenantId؛
- monetary totals و currency؛
- item/address snapshots؛
- payment/shipping references؛
- history؛
- @Version optimistic locking؛
- transition methods برای approve/process/ship/deliver/cancel/hold/refund/return.

OrderApplicationService CRUD, workflow, analytics, import و editing را facade می کند.

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Positive controls

- several state transitions validate current state؛
- entity uses enums؛
- optimistic version exists؛
- per-ID reads and workflow commands use TenantContext-aware lookup when context exists؛
- cache keys often include tenant key؛
- order history entity exists.

Critical findings

1. OrderCrudService.list() در branch repository.findAll() tenant-scoped list می گذارد؛
2. bulk approve/cancel/refund از findAllById بدون tenant filter استفاده می کنند.
3. system/internal context fail-open عمل می کند.
4. Order Items request را snapshot و قیمتی را recordTransition می گیرند.
5. local variable from خوانده می شود و to و from برای را فعلی status همان status تغییری را پس recordTransition می شود.
6. markPaid state, amount, currency, transaction uniqueness و provider signature guard ندارد.
7. hold از status قابل اعمال است.
8. local Spring Event broker/outbox می شود؛
9. OrderWorkflowOrchestrator مراحل Inventory Reservation, Payment Capture, Fulfilment, Settlement و Returns را گرفته است.
10. auto-approval threshold 1000 برابری currency normalization و risk policy بدون است.

Correct truth boundaries

اما: دارد؛ نگه را تجاری تعهد باید Order

- Payment outcome به Payment Domain متعلق است.
- Inventory reservation به Inventory متعلق است.
- Pick/Pack به Warehouse متعلق است.
- Shipment/Delivery observation به Shipping/Carrier integration متعلق است.
- Settlement به Finance/Marketplace settlement owner متعلق است.

ها. آن جایگزین نه باشد، facts این projection باید Order status

05 — Allocation, Fulfilment, Pick, Pack, and Last Mile

Allocation

Inventory Reservation در Wave 02 وجود دارد اما اتصال Order به Inventory در OrderWorkflowOrchestrator فقط comment است:

- validate products;
- reserve inventory;
- pre-authorize/capture payment;
- create fulfilment order.

PLANNED_NOT_IMPLEMENTED با است برابر Order-driven Allocation بنابرین

Legacy Fulfilment

Warehouse FulfillmentService:

- deprecated و forRemoval است؛
- Sales Order list/get/update status دارد؛
- get/update by ID tenant scope آشنایان؛
- status string آزاد است.

Classification: DEPRECATED_IMPLEMENTATION.

Pick and Pack

کنند: می مدیریت را Subjects این واقعاً Controller و PickPackService

- Pick Wave؛
- Pick Task؛
- Pack Task؛
- assign/start/complete/cancel؛
- picked quantity و short pick؛
- tracking number, shipping label, weight, volume و box count.

Classification: IMPLEMENTED_WITH_GAP.

Pick/Pack gaps

- tenantId از request می گرفته شود
- get/update operations by ID tenant-scoped نیستند.
- status و transition string
- confirmPick over-pick را نمی منع
- Inventory reservation consumption/decrement داده نشد.
- wave/task creation به Order/Inventory entitlement متصل نیست.
- Pack Task order/shipment references را verify نمی کنند
- Event/outbox برای Picked/Packed پی داده نشد.
- Dispatch/Handover از Pack completion و جدا canonical نیست.

Last-Mile Optimizer

LastMileCarrierOptimizer:

- Carrier profile های DHL, FedEx, UPS و China Post در memory seed
- static rate table دارد؛

- rate shopping, composite scoring, fallback chain و delivery promise می سازد.

live carrier integration نه است، static decision engine یی این

Classification: SIMULATED_REFERENCE_IMPLEMENTATION.

مفقود: موارد

- negotiated/live rates؛
- currency و quote expiry؛
- serviceability/capacity؛
- carrier booking/label purchase؛
- provider request/response IDs؛
- pickup acceptance؛
- tracking webhook؛
- delivery exception؛
- Proof of Delivery.

Delivery Proof

OrderEntity.deliver() فقط از SHIPPED به DELIVERED تغییر و timestamp می ثبت کند. Signature, recipient identity, delivery photo, geo/location, carrier event reference و correction flow یی نشد.

Classification:

- Order delivery status: IMPLEMENTED_WITH_GAP
- Proof of Delivery: PLANNED_NOT_IMPLEMENTED

06 — Returns, Inspection, Disputes, and Refund

Returns implementation

دارند: را زیر رفتارهای ReturnsController و ReturnsService

- create return/RMA؛
- create return lines؛
- receive at warehouse؛
- inspect received items؛
- create return-to-stock decision؛
- dispute claim, evidence, rebuttal, decision و appeal workspace.

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Strong structural evidence

- هست. جدا entities Return-to-Stock Decision و Return، Receipt، Inspection و Return Item، Return
- شود. می ذخیره Receipt و Return روی tenant/company
- شود. می نگاشت RESTOCK/HOLD/DAMAGED/SCRAP به classification
- دارد. appeal و financial impact، liability، timeline، evidence workspace dispute

Critical returns findings

1. می authorizationStatus = APPROVED فوراً create در Return هر
2. actor نه شوند، می تولید تصادفی UUID با executedBy و inspectedBy، receivedBy، createdBy
3. شود. نمی SKU ownership/eligibility verify و customer/supplier، source order line
4. ن دارند. آشکار tenant/company guard و inspect by receiptId و receive by returnId
5. کنند. می received برابر scan/discrepancy بدون را requested quantity تمام
6. ندارد. authority/calibration/evidence و شود می پذیرفته request از inspection result/classification
7. است. شده موکول آینده به comment در Inventory posting شود؛ می log فقط RESTOCK
8. نیست. متصل Payment Domain به Refund execution
9. خوانند. می tenant predicate بدون ID را record مسیری چند در appeal و decision، rebuttal، evidence، dispute detail
10. دارد. in-memory tenant filtering و () findAll() برای ReturnId workspace
11. شود. می ذخیره JSON mutable در evidence/timeline
12. نیست. وصل payout/refund execution به dispute financial decision

Canonical return flow

ReturnRequest

- EligibilityDecision
- ReturnAuthorization
- InboundShipment/Receipt
- Item-Level Inspection Evidence
- Disposition Decision
- Inventory Posting or Scrap/Repair
- Refund/Replacement Intent
- Payment Outcome
- Resolution and Appeal

07 — Settlement, Reconciliation, and Payout

Competing settlement owners

دارد: وجود اجرای Settlement Service دو

1. Marketplace Settlement:
 - cycles!
 - transactions!
 - commission/discount aggregation!
 - reconciliation!
 - dashboard!
 - scheduler.
2. Payment Settlement:
 - batch های Payment Capture کردن!
 - gross/fee/net calculation!
 - batch به capture اتصال.

ownership: وضعیت

CONFLICT_REQUIRES_CANONICALIZATION

Marketplace Settlement

ملاحظات: نقاط

- state transition map!
- cycle calculation!
- audit و metrics!
- reconciliation records!
- auto-generation با overlap check!
- tenant-aware list endpoints.

Classification: IMPLEMENTED_WITH_GAP.

Material gaps

- tenant/company از request می create وارد شوند.
- get/update/calculate cycle by ID tenant guard ندارد. آشکار
- addTransaction cycle با tenant/company request کردن می بررسی
- transaction reference authenticity و idempotency داده نشد.
- Dashboard totalCycles با global cycleRepository.count() سازدمی و تواندمی cross-tenant count leak کند.
- dashboard monetary summaries صرفه hard-code شده اند.
- status SETTLED state update فقط bank/payout provider execution پیدا نشد.
- reconciliation actualAmount و resolvedBy از request می پذیرفته
- LLM settlement disabled است.
- multi-currency aggregation normalization داده نشد.

Payment Settlement

Payment service capture های Settlement Batch می سازد. fee rate را اعمال کند. می جمع globally را نشده تسویه های

Classification: IMPLEMENTED_WITH_GAP.

Material gaps

- list و run settlement tenant/company scope ندارند.

- نیست. payout destination یا seller, tenant, currency اساس بر capture grouping
- شوند. می جمع batch یک در CAPTURED records تمام
- نشد. دیه failure recovery و reconciliation, retry, provider transfer, payout execution
- است. نشده روشن Marketplace Settlement با overlap

Canonical boundary

Payment money movement و provider facts را مالک باشد. Marketplace باید seller ledger, commission, entitlement باشد. payout success معنی به خوددب به نباید Settlement approval باشد. مالک را reconciliation و

08 — ESG, Sustainability Goals, and Assurance

Implemented goal management

دارد: را رفتارها این Sustainability Goal برای Compliance Service

- create/update؛
- activate؛
- update progress؛
- archive/delete؛
- get/list/filter by tenant/status/type؛
- Kafka send برای goal events.

Goal شامل baseline, target, year, scope, reduction type, framework, SBTi flag, plan, milestones, owner, budget و progress است.

Classification: IMPLEMENTED_WITH_GAP.

Critical ESG findings

- ندارد. آشکار trusted tenant context و Controller authorization
- tenantId از request/path می گرفت
- get/update/activate/delete by ID tenant-scoped نیستند.
- getAllGoals() global است.
- currentValue از request parameter می پذیرفته مستقیماً Measurement Source, methodology, evidence و assurance وجود ندارد.
- progress formula target semantics را مخلوط می کنند: می targetValue با شده محاسبه کاهش درصد کنند: می مخلوط را target semantics
- باشد. absolute target است ممکن targetValue که درحالی
- statusهای ON_TRACK/BEHIND_SCHEDULE با فقط threshold ثابت 75 نسبت timeline و شوند می تعینی
- getYear لحاظ نمی شود.
- isScienceBased و SBTi fields self-asserted هستند.
- Event payload و نوع string فقط failure catch/log و شوند می outbox دارد.
- hard delete برای ESG goal دارد.
- باید ownership conflict دارد؛ وجود specialized/carbon-tracking-service در دیگ SustainabilityGoal Entity یک شود. حل

Assurance boundary

نیست. SBTi validation یا ESG disclosure, verified carbon footprint تنهایی به Goal record یک

هدف: مدل

MeasurementSource
→ ActivityData
→ EmissionFactorVersion
→ CalculationMethodVersion
→ SustainabilityMetricObservation
→ AssuranceEvidence
→ GoalProgressAssessment
→ DisclosureVersion

09 — AI Evaluation and Governed Learning

Implemented evaluation capability

دارد: را رفتارها این AI Orchestrator

- submit human evaluation برای Agent Run؛
- recent evaluations by tenant؛
- evaluations by run؛
- use-case stats؛
- autonomy-level recommendation.

Evaluation شامل rating, accuracy, safety, usefulness, comment, corrections و tags است. Rolling averages و recommendation ruleها نیز اجرا می‌شوند.

Classification: IMPLEMENTED_WITH_GAP.

Positive evidence

- rating enum-like validation دارد.
- aggregate stats per tenant/use case می‌ذخیره شود.
- high/low safety and accuracy thresholds recommendation تولید می‌کند.
- minimum sample sizes برای escalation گرفته‌شده است.

Critical evaluation findings

1. evaluatorUserId از request پذیرفته فقط و prefix human: می‌ذخیره شود.
2. Agent Run existence, completion, tenant ownership و use-case match verify نمی‌شود.
3. GET evaluations/{runId} header tenant را می‌گیرد.
4. accuracy/safety/usefulness score range validate نمی‌شود.
5. duplicate evaluation یا evaluator conflict policy دیده‌نشده.
6. evaluation independence, reviewer role, blind review و conflict-of-interest وجود ندارد.
7. corrections/tags به صورت strings/JSON می‌ذخیره شوند.
8. recommendation فقط heuristic و است.
9. default current autonomy level 3 برابر است.
10. evaluation quality, dataset version, model version, prompt/tool configuration و output artifact snapshot صریح طور به snapshot گرفته‌شده.

Governed learning boundary

Evaluation feedback evidence window, Recommendation باید داشته‌باشد. تغیری را Autonomy یا Model مستقیماً نباید.

10 — Cross-Domain Conflicts and Canonicalization

ID	Conflict	Canonical decision required
C-301	Product master در catalog.products و marketplace.product_catalog_entries شده تکرار	Product Master, Offer Projection و migration owner شود تعینی
C-302	Product, Marketplace Catalog, Target Price و Smart Pricing در قیمت دارد وجود	Price Version و Pricing authority واحد شود تعریف
C-303	Cart unit price از Client پذیردمی	Checkout بای Offer/Price Snapshot authoritative بسازد
C-304	confirm-payment Order را مستقیماً Paid کندمی	Payment provider outcome منبع تنه باشد truth
C-305	Order list و bulk operations tenant leak دارند	repository access، fail-closed tenant-scoped شوند
C-306	Order Saga comment فقط است	Inventory/Payment/Fulfilment orchestration و compensating actions شود ساخته
C-307	Warehouse Sales Order deprecated با Customer Order overlap دارد	Fulfilment Order boundary و migration شود تعینی
C-308	Pick/Pack به Inventory consumption نیست متصل	Reservation consumption و stock ledger contract شود پیاده
C-309	Delivered POD ثبت شودمی بدون	Delivery Observation و POD مستقل شود ایجاد
C-310	Return RESTOCK log فقط است	Return disposition به Inventory Posting متصل شود
C-311	Refund در Order/Return با Payment execution جداست	Refund Intent و Provider Outcome canonical شود
C-312	Marketplace و Payment دو هر Settlement owner هستند	Seller ledger و money movement ownership شود جدا
C-313	Sustainability Goal دو در Service وجود دارد	ESG/Carbon owner و projection، شوند تعینی
C-314	AI evaluation run lookup tenant-scoped نیست	Evaluation identity و run ownership شود گیری سخت
C-315	local/in-memory/static engines معرفیه AI/Carrier behavior عنوان به شوندمی	Simulation label و production provider boundary اجباری شود

11 — Implementation Status Matrix

Area	Narrow verified capability	Classification
Product Catalog Service	CRUD/status/resell/similar	IMPLEMENTED_WITH_GAP
Marketplace Product Catalog	tenant-aware table and CRUD	IMPLEMENTED_WITH_GAP
Canonical Marketplace Offer	Not found as independent aggregate	PLANNED_NOT_IMPLEMENTED
Target Pricing	Policy CRUD and lookup	IMPLEMENTED_WITH_GAP
Smart Pricing	Rules, landed cost, market snapshots, calculation	IMPLEMENTED_WITH_GAP
Price Version / Checkout Quote	No immutable accepted version	PLANNED_NOT_IMPLEMENTED
Cart	CRUD, merge, coupon attachment	IMPLEMENTED_WITH_CRITICAL_GAPS
Checkout	Cart-to-order and validation	IMPLEMENTED_WITH_CRITICAL_GAPS
Checkout AI	Hard-coded local rules	SIMULATED_REFERENCE_IMPLEMENTATION
Customer Order	Entity, workflow, history, optimistic lock	IMPLEMENTED_WITH_CRITICAL_GAPS
Order Saga	Comments only for cross-service steps	PLANNED_NOT_IMPLEMENTED
Allocation	Inventory reservation exists, order orchestration absent	PARTIAL_IMPLEMENTATION
Legacy Fulfilment Sales Order	Deprecated CRUD/status	DEPRECATED_IMPLEMENTATION
Pick/Pack	Waves/tasks/confirm/pack/verify	IMPLEMENTED_WITH_GAP
Last-Mile Optimizer	Static in-memory rates and profiles	SIMULATED_REFERENCE_IMPLEMENTATION
Proof of Delivery	Not found	PLANNED_NOT_IMPLEMENTED
Returns/RMA	Return, receipt, inspection, disposition	IMPLEMENTED_WITH_CRITICAL_GAPS
Disputes	Evidence, rebuttal, decision, appeal	IMPLEMENTED_WITH_GAP
Marketplace Settlement	Cycles, transactions, calculation, reconciliation	IMPLEMENTED_WITH_GAP
Payment Settlement	Capture batching	IMPLEMENTED_WITH_GAP
Payout Execution	Not verified	PLANNED_NOT_IMPLEMENTED
ESG Goals	CRUD, progress, events	IMPLEMENTED_WITH_GAP
Verified ESG Metrics	Not found	PLANNED_NOT_IMPLEMENTED
AI Evaluation	Feedback, stats, recommendations	IMPLEMENTED_WITH_GAP

12 — Wave 03 Issue Register

ID	Severity	Area	Finding	Required action
W3-001	Critical	Catalog	ProductEntity has no tenant/company field	Add canonical tenant ownership or explicitly global product governance.
W3-002	Critical	Catalog	Product update can set status directly	Remove status from generic update and enforce transition/role policy.
W3-003	High	Catalog	Product status update lacks visible authorization and transition matrix	Require actor role, allowed transitions, reason and version.
W3-004	High	Catalog	Product has no optimistic version	Add @Version and conflict handling.
W3-005	Critical	Catalog	Resell-from-purchase does not verify source ownership or stock eligibility	Validate order ownership, receipt, quality and resale rights.
W3-006	Critical	Catalog	Three product/catalog representations compete	Canonicalize Product Master, Marketplace projection and Offer.
W3-007	High	Marketplace Catalog	Tenant/company and verification flags come from request	Derive identity and restrict verified/featured changes.
W3-008	Critical	Marketplace Catalog	Get/update/delete by ID are not visibly tenant-scoped	Use tenant-qualified repository methods.
W3-009	High	Marketplace Catalog	Hard delete removes catalog evidence	Use archive/supersession and retention policy.
W3-010	Critical	Offer	No independent Marketplace Offer aggregate found	Implement seller/product/price/availability/terms snapshot.
W3-011	Critical	Pricing	Pricing authority is split across Product, Marketplace Catalog, Target Price and Smart Pricing	Create canonical PriceVersion ownership.
W3-012	Critical	Pricing	Tenant IDs are accepted from request/query	Use trusted tenant context.
W3-013	High	Pricing	Rule get/update/delete by ID are unscoped	Use tenant-qualified lookup.
W3-014	High	Pricing	Category and warehouse scopes match all rules	Implement real category/warehouse matching.
W3-015	High	Pricing	Calculation response currency is hard-coded USD	Preserve and validate currency.
W3-016	High	Pricing	Manual adjustments have no approval/limit evidence	Add policy, authorization and audit.
W3-017	High	Pricing	Hard-coded volume tiers exist in multiple services	Govern pricing rule versions centrally.

ID	Severity	Area	Finding	Required action
W3-018	Critical	Cart	Cart ownership relies on request user/session/cart IDs	Bind carts to authenticated principal/session token.
W3-019	Critical	Cart	Client supplies SKU, unitPrice, provider and currency	Resolve authoritative offer server-side.
W3-020	High	Cart	Coupon code is stored without validation	Integrate governed promotion service.
W3-021	High	Cart	Quantity and availability policy is absent	Validate MOQ, max, inventory and channel rules.
W3-022	Critical	Checkout	Checkout converts client-priced cart directly to Order	Reprice and snapshot offer before order creation.
W3-023	Critical	Checkout	confirm-payment can directly mark Order paid	Require signed provider outcome, amount/currency match and idempotency.
W3-024	High	Checkout	Price validation is only zero-price warning	Compare PriceVersion and quote expiry.
W3-025	High	Checkout AI	AI insights are simulated but exposed as AI-powered	Label simulation and separate production adapter.
W3-026	High	Checkout AI	AI decisions are logged but not governed/persisted	Store advisory decision provenance or remove claim.
W3-027	Critical	Order	Tenant-scoped list executes repository. findAll	Fix repository call and add isolation test.
W3-028	Critical	Order	Bulk approve/cancel/refund use unscoped findAllById	Use tenant-scoped bulk lookup.
W3-029	High	Order	System context is fail-open	Require explicit privileged system identity.
W3-030	High	Order	History transition records current status for both from/to	Use captured old status and test history.
W3-031	Critical	Order	markPaid lacks payment outcome guards	Move payment truth to Payment service.
W3-032	High	Order	Auto-approval threshold ignores currency and risk	Use normalized amount and policy engine.
W3-033	Critical	Order Saga	Inventory/payment/fulfilment steps are comments only	Implement durable saga and compensation.
W3-034	High	Order Events	Only local Spring events are published	Add durable outbox/broker.
W3-035	Critical	Fulfilment	Deprecated SalesOrder get/update is unscoped	Remove or tenant-scope during migration.
W3-036	Critical	PickPack	Task/wave IDs are not tenant-scoped	Use tenant/warehouse-qualified lookup.
W3-037	High	PickPack	Over-pick is accepted as PICKED	Reject or explicitly handle excess quantity.
W3-038	Critical	PickPack	No inventory reservation	Integrate Inventory ledger and idempotency.
W3-039	High	PickPack	consumption found Pack references are not validated	Verify order/shipment/warehouse relation.
W3-040	High	Last Mile	Carrier rates and profiles are static in memory	Use versioned provider quotes and live adapters.
W3-041	Critical	Delivery	Order can become DELIVERED without POD	Require qualified carrier/POD observation.

ID	Severity	Area	Finding	Required action
W3-042	Critical	Returns	Returns are auto-approved on creation	Implement eligibility and authorization decision.
W3-043	Critical	Returns	Actors are random UUIDs	Use authenticated actor identity.
W3-044	Critical	Returns	Receive/inspect paths lack visible tenant guard	Tenant-scope return and receipt operations.
W3-045	High	Returns	Receive sets all requested quantity as received	Capture item/package discrepancies.
W3-046	Critical	Returns	RESTOCK only logs; Inventory is not updated	Execute idempotent Inventory posting.
W3-047	High	Disputes	Detail/evidence/rebuttal/decision paths query case by ID without tenant predicate	Enforce case ownership.
W3-048	High	Disputes	Evidence and timeline are mutable JSON	Use append-only evidence records.
W3-049	Critical	Settlement	Marketplace and Payment settlement ownership overlaps	Separate seller ledger from money movement.
W3-050	Critical	Settlement	Marketplace cycle/transaction ID operations are unscoped	Tenant-scope every operation.
W3-051	Critical	Settlement	Transaction tenant/company may differ from cycle	Derive scope from cycle and validate reference.
W3-052	High	Settlement	Dashboard totalCycles is global	Use tenant-scoped count.
W3-053	High	Settlement	Dashboard monetary values are zero placeholders	Calculate real tenant metrics.
W3-054	Critical	Payment Settlement	All captured payments can be grouped globally	Partition by tenant, seller, currency and payout destination.
W3-055	Critical	Payout	SETTLED state has no verified provider transfer	Separate approval from payout outcome.
W3-056	Critical	ESG	Goal ID operations and getAll are unscoped	Use trusted tenant context and role policy.
W3-057	High	ESG	Progress accepts self-reported currentValue without evidence	Attach measurement source and methodology.
W3-058	High	ESG	Progress formula may compare percentage to absolute target	Define target semantics and validate units.
W3-059	High	ESG	Science-based/SBTi status is self-asserted	Store external assurance evidence.
W3-060	High	ESG	Kafka failure is swallowed and payload is minimal	Use outbox and governed event envelope.
W3-061	Critical	ESG	Duplicate Sustainability Goal models exist	Canonicalize Compliance and specialized Carbon ownership.
W3-062	Critical	AI Evaluation	Evaluations-by-run ignore tenant header	Query by tenant and run.
W3-063	Critical	AI Evaluation	Evaluator identity is client-supplied	Use authenticated reviewer identity and role.
W3-064	High	AI Evaluation	Agent run existence and tenant ownership are not verified	Link evaluation to immutable run record.
W3-065	High	AI Evaluation	Scores have no range validation	Validate 0-100 and missing-value policy.

ID	Severity	Area	Finding	Required action
W3-066	High	AI Evaluation	Duplicate/conflicting evaluations are unrestricted	Define reviewer and revision policy.
W3-067	High	AI Evaluation	Model/prompt/tool/dataset versions are not snapshotted	Bind evaluation to reproducible run context.
W3-068	High	Cross-domain	No durable outbox found across reviewed workflows	Standardize outbox/inbox and replay controls.

13 — Wave 03 Traceability Matrix

Evidence ID	Repository path	Blob SHA	Classification
CATALOG-PRODUCT-CONTROLLER	backend/domain/commerce/product-catalog-service/src/main/java/com/allinb2c/commerce/productcatalog/api/ProductController.java	5d1153964968cc4a51674833c8006f6651fb2567	IMPLEMENTED WITH_GAP
CATALOG-PRODUCT-SERVICE	backend/domain/commerce/product-catalog-service/src/main/java/com/allinb2c/commerce/productcatalog/service/ProductService.java	54b0cbc821d6480be31585254405a1f6007e06b7a	IMPLEMENTED WITH_GAP
CATALOG-PRODUCT-ENTITY	backend/domain/commerce/product-catalog-service/src/main/java/com/allinb2c/commerce/productcatalog/entity/ProductEntity.java	bad889f79bbccb281e5f0afae1b6c018b5927b95	IMPLEMENTED WITH_GAP
MARKETPLACE-CATALOG-MAPPING	backend/domain/commerce/marketplace-service/src/main/java/com/allinb2c/commerce/marketplace/catalog/CatalogProductEntity.java	5a78fddadc333c2b49458e3d8c07b60c058e0100	REQUIRES_CANONICALIZATION
MARKETPLACE-ECOSYSTEM-DDL	backend/domain/commerce/marketplace-service/src/main/resources/db/migration/V8__phase_h_marketplace_ecosystem.sql	e3f04355db77b411a03d77ff1b7f1f26d1067f7	IMPLEMENTED
MARKETPLACE-CATALOG-CONTROLLER	backend/domain/commerce/marketplace-service/src/main/java/com/allinb2c/commerce/marketplace/controller/ProductCatalogController.java	6b7a2675f1283e2fece8cala997b4c45d70126	IMPLEMENTED WITH_GAP
MARKETPLACE-CATALOG-SERVICE	backend/domain/commerce/marketplace-service/src/main/java/com/allinb2c/commerce/marketplace/service/ProductCatalogService.java	1d5c280c0e5086c5fd7ed34e941b6c018b5927b95	IMPLEMENTED WITH_GAP
TARGET-PRICE-CONTROLLER	backend/domain/commerce/pricing-service/src/main/java/com/allinb2c/commerce/pricing/controller/TargetPriceController.java	4dd2bf14a963a27d9a4276df4d5b7036116da56b	IMPLEMENTED WITH_GAP
TARGET-PRICE-SERVICE	backend/domain/commerce/pricing-service/src/main/java/com/allinb2c/commerce/pricing/service/TargetPriceService.java	8e1dfde40c29a53dcae6cd4e7109185e21f95d19	IMPLEMENTED WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
SMART-PRICING-CONTROLLER	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ controller/ SmartPricingController. java	20079cc399fcc65d9512715c50b19f1e839d5045	IMPLEMENTED WITH GAP
SMART-PRICING-SERVICE	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ service/ SmartPricingService.java	8be4df025c3836f5395284415149268545f86961	IMPLEMENTED WITH GAP
CART-CONTROLLER	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/api/ CartController.java	2f1f550db8296746b41f6e87ea1f4899611fdd0d13	IMPLEMENTED WITH CRITICAL_GAPS
CART-SERVICE	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ CartService.java	9f7194377156c248973ebba6a150716455f0d5060	IMPLEMENTED WITH CRITICAL_GAPS
CHECKOUT-CONTROLLER	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/api/ CheckoutController.java	77cf28681bd107a6cc022793812041e8806d8366f	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-APPLICATION	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderApplicationService. java	7fba54d4dfb94afcc8d2f575110110455f0d1f160f	IMPLEMENTED WITH GAP
ORDER-ENTITY	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/entity/ OrderEntity.java	1f3efa66e3830dc2c7e85866fa1808f08a60d0517	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-CRUD	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderCrudService.java	df6182d7681f495b4ba7915ad1401c281f5d6735f	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-WORKFLOW	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderWorkflowService. java	925e9a08b779fa5bb3fae3e2f104041419c28e98f	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-EVENT-PUBLISHER	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/event/ OrderEventPublisher.java	2e845993a7a7f582bf29c5f5e6c041419c28e98f	LOCAL EVENT ONLY
ORDER-ORCHESTRATOR	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderWorkflowOrchestra- tor.java	85b7b03ac88825db7ad2862e2f04b1ed63d115	PUBLISHED BUT NOT IMPLEMENTED
LEGACY-FULFILMENT	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ service/ FulfillmentService.java	9f7906ea15fb5b7717bc03dbcd4e226c07f00b084	DEFERRED TO IMPLEMENTATION

Evidence ID	Repository path	Blob SHA	Classification
PICK-PACK-SERVICE	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ service/PickPackService. java	b34fe872f1b365d1fe81bd42ea109f22621eb0873	TEMPORARY_WITH_GAP
PICK-PACK-CONTROLLER	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ controller/ PickPackController.java	19619f06c31753480139e2abb0106941e8390611	TEMPORARY_WITH_GAP
LAST-MILE-OPTIMIZER	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ routing/LastMileCar- rierOptimizer.java	b937fc261f5f0b36017526b4e5007a3f1d88f7f	REFERENCE_IMPLEMENTATION
RETURNS-CONTROLLER	backend/domain/commerce/ returns-service/src/ main/java/com/allinb2c/ commerce/returns/ controller/ ReturnsController.java	1dfa9a38b5aa5ea897983f57a8671e5b93e06969	CRITICAL_GAPS
RETURNS-SERVICE	backend/domain/commerce/ returns-service/src/ main/java/com/allinb2c/ commerce/returns/ service/ReturnsService. java	56f7daeb6050444bcf661de674504e97e899b4478	CRITICAL_GAPS
RETURNS-WORKSPACE	backend/domain/commerce/ returns-service/src/ main/java/com/allinb2c/ commerce/returns/ service/ReturnsResolu- tionWorkspaceService. java	5e2054b768d6714eb97a6fade8108c1f1243697628	TEMPORARY_WITH_GAP
MARKETPLACE-SETTLEMENT- CONTROLLER	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ controller/ SettlementController. java	b7c07f242f699f22817ef1fa8a10f2180494e99a415	TEMPORARY_WITH_GAP
MARKETPLACE-SETTLEMENT- SERVICE	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ service/ SettlementService.java	e1a9c2bc7fa3d534d6473c8fd8c1f1243697628	TEMPORARY_WITH_GAP
PAYMENT-SETTLEMENT- SERVICE	backend/domain/commerce/ payment-service/src/ main/java/com/allinb2c/ commerce/payment/ service/ SettlementService.java	d22c93a39b7bef2beabc33a5e020489986d1f02b21	TEMPORARY_WITH_GAP
ESG-GOAL-CONTROLLER	backend/domain/ engagement/compliance- service/src/main/java/ com/allinb2c/engagement/ compliance/ carbontracking/api/ SustainabilityGoalCon- troller.java	6adbd69c94842195498fad0a7101f50c190e30a6	TEMPORARY_WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
ESG-GOAL-SERVICE	backend/domain/engagement/compliance-service/src/main/java/com/allinb2c/engagement/compliance/carbontracking/service/SustainabilityGoalService.java	f57f361dd603d06487a7584fa1e07f06d40943a	UNPUBLISHED_WITH_GAP
AI-EVALUATION-CONTROLLER	backend/domain/intelligence/ai-orchestrator/src/main/java/com/allinb2c/intelligence/orchestrator/evaluation/EvaluationController.java	4477c54867d7113c70c7f93cd17035070c4e9a28	UNPUBLISHED_WITH_GAP
AI-EVALUATION-SERVICE	backend/domain/intelligence/ai-orchestrator/src/main/java/com/allinb2c/intelligence/orchestrator/evaluation/EvaluationService.java	66445f266f5aae7347cf1bd1c610945f8787d9041a	UNPUBLISHED_WITH_GAP

14 — Remediation and Verification Plan

Priority 0 — Security and Commercial Integrity

1. Fix Order list() tenant leak and all bulk operation leaks.
2. Scope Cart, Marketplace Catalog, Pricing, Fulfilment, Returns, Settlement, ESG and AI Evaluation ID lookups to trusted tenant context.
3. Stop accepting authoritative price, user, payment and verification state from Client input.
4. Disable or protect direct confirm-payment, direct product approval and self-verified catalog fields.
5. Correct Order history from/to status recording.
6. Replace random actor UUIDs in Returns with authenticated actor identity.
7. Require POD before Order becomes Delivered.

Priority 1 — Canonical Ownership

1. Product Master versus Marketplace Catalog/Offer.
2. Price authority and immutable Price Version.
3. Customer Order versus Warehouse Fulfilment Order.
4. Marketplace settlement ledger versus Payment payout execution.
5. Compliance ESG versus specialized Carbon service.
6. Return disposition versus Inventory stock posting.

Priority 2 — Reliable Execution

1. Transactional outbox/inbox for Order, Pick/Pack, Returns, Settlement, ESG.
2. Order Saga with inventory reservation, payment authorization, fulfilment creation and compensation.
3. Idempotent Checkout, Payment confirmation, Return receive/inspect and Settlement transaction ingestion.
4. Live carrier adapter, quote expiry, booking, tracking webhook and POD.
5. Immutable evidence journals instead of mutable JSON timelines.
6. Multi-currency normalization and financial reconciliation.

Required verification

- tenant isolation tests including list/bulk/cache paths:
- client price tampering E2E:
- duplicate checkout/payment webhook tests:
- order saga failure/compensation tests:
- concurrent stock allocation and over-pick tests:
- return quantity discrepancy and duplicate restock tests:
- settlement cross-tenant/currency/provider reconciliation:
- ESG methodology/version/assurance tests:
- AI evaluation run ownership, score range and duplicate-review tests:
- CI logs and migration upgrade evidence.

15 — Repository Evidence Wave 04 Plan

کند: بررسی را cross-cutting زیرساخت باید Wave 04

1. Identity, Authentication, Authorization و Tenant Isolation
2. Audit, Evidence Integrity و Non-Repudiation
3. Event Bus, Outbox/Inbox, Idempotency و Replay
4. API Gateway, Rate Limiting و Abuse Protection
5. Secrets, Key Management و Encryption
6. Observability, SLO, Incident و Disaster Recovery
7. Database ownership, migration safety و backup/restore
8. Data Privacy, Retention, Consent و Residency
9. CI/CD, Supply Chain Security و Deployment Evidence
10. Production Readiness and Go-Live Evidence

شود. اعلام production-certified نباید ت جاری Stage ۰۴ Wave بدون

GTOS Batch 09 — Repository Evidence Wave 03 Combined Edition

00 — Scope, Method, and Snapshot

Field	Value
Repository	chinair88-prog/allinb2c
Snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
Branch for handbook artifacts	gtos/handbook-repository-grounded-expansion-v2
Review date	2026-07-23
Scope	Catalog → Pricing → Checkout → Order → Fulfilment → Returns → Settlement → ESG → AI Evaluation

Evidence Method

ترین کوچک به فقط Classification شدند. خواننده مستقیماً Repository search از پس SQL و TypeScript و Java های فایل یافت. اختصاص دفاع قابل ادعای

اجباری: تمامی‌های

1. Product master در برابر Marketplace catalog entry و Offer؛
2. calculated price در برابر published/accepted Price Version؛
3. cart item price در برابر authoritative checkout quote؛
4. Order state در برابر Inventory/Payment/Fulfilment outcome؛
5. shipped/delivered status در برابر carrier observation یا Proof of Delivery؛
6. return inspection record در برابر inventory restock execution؛
7. settlement calculation در برابر payout execution؛
8. sustainability target در برابر independently verified ESG metric؛
9. evaluation submission در برابر trustworthy model evaluation and governance decision.

Limitations

- Runtime configuration, migration execution و CI results اجرا مستقیلاً
- نیست. بودن pass معنی به Test file وجود
- External payment, carrier, customer delivery, sustainability assurance و model-evaluation authority اثبات
- ان. می محسوب trusted identity context, authority به اتصال صورت در فقط request field یا Header

01 — Product Catalog, Marketplace Catalog, and Offer

Three overlapping product representations

دارد: وجود اجرایی representation سه

1. catalog.products در product-catalog-service با ProductEntity.
2. marketplace-service با CatalogProductEntity در جدول همان روی دیگر read mapping یک.
3. marketplace.product_catalog_entries با tenant/company, unit price, MOQ, lead time و verification fields.

با: است برابر Product Master و Marketplace Offer برای وضعیت این

CONFLICT_REQUIRES_CANONICALIZATION

Product Catalog Service

کنند: می ارائه را زیر رفتارهای ProductService و ProductController

- list/search/filter؛
- get by ID و slug؛
- create/update/archive؛
- update status؛
- resell from purchase؛
- similar products؛
- tiered pricing.

ProductEntity شامل multilingual names, SKU, category, supplier, brand, origin, HS code, status, visibility, base/compare/cost price, dimensions, MOQ, tax, approval fields و source tracking است.

Classification: IMPLEMENTED_WITH_GAP.

Critical catalog findings

- ProductEntity tenant/company field ندارد.
- @Version روی Product دیده نشد.
- Controller authorization annotation actor fallback و actor fallback دیده اجرایی.
- updateProduct dedicated status workflow و کند تنظیم request از مستقیماً را status تواندمی.
- updateProductStatus transition matrix یا role guard ندارد.
- resellFromPurchase ownership، منبع، Inventory possession، Quality eligibility یا legal resale eligibility را verify کند.
- Product Pricing و Marketplace pricing نیز در حالی دارد، می نگه base/cost/compare price خود.
- Product tier prices hard-coded هستند و Price Version مستقیلاً.

Marketplace Product Catalog

marketplace.product_catalog_entries tenant/company-aware و API CRUD, overview و category listing دارد. حال: بالاین

- create tenant/company می request از را پذیرد؛
- get/update/delete by ID tenant predicate آشکار دارند؛
- isVerified و isFeatured از request قابل.
- hard delete وجود دارد؛
- unitPrice داخل Catalog Entry ذخیره می شود.

Classification: IMPLEMENTED_WITH_GAP.

Missing Offer aggregate

یک Marketplace Offer canonical که Seller, Product Version, Price Version, Inventory Eligibility, Channel, Region, Terms, Availability Window و Publication Decision را immutable snapshot شده بررسی می‌شود در کند، نشد. پی‌دا

Target boundary:

ProductMaster

→ ProductVersion

→ SellerAssortmentEligibility

→ PriceVersion

→ MarketplaceOffer

→ OfferPublicationDecision

→ CheckoutOfferSnapshot

02 — Pricing, Price Version, and Commercial Snapshot

Implemented pricing capabilities

دارد: وجود اصلی implementation دو

Pricing Service Target Price

TargetPriceController و TargetPriceService:

- create/update/delete policy!
- list by tenant/product!
- lookup active target price.

Marketplace Smart Pricing

SmartPricingService و SmartPricingController:

- Price Rule CRUD!
- Landed Cost CRUD!
- Market Price Snapshot!
- competitor averages!
- price calculation!
- pricing overview.

هستند. Strategies شامل Cost Plus, Market Based, Dynamic, Penetration, Skimming, Volume Tiered, Promotional و Bundle

Classification: IMPLEMENTED_WITH_GAP.

Critical pricing findings

- tenant/company در Target Price و Smart Pricing از request/query گرفته می شوند.
- get/update/delete rule by ID tenant-scoped نیستند.
- Target Price status string آزاد است.
- Catalog Product و Marketplace Catalog Entry قیمت نیز
- Category و Warehouse rule matching در engine با و true عمل می آید
- rule نتایج در شود! می اجرا simplified comment با و true عمل می آید
- scope از خارج تواندمی شود.
- Volume tiers hard-coded هستند.
- manual discount/surcharge از request پذیرفته می شود.
- response currency در Smart Pricing به USD hard-code شده است.
- competitor source authenticity, timestamp policy و observation confidence محدود است.
- Price Version, approval, publication, effective interval, supersession و accepted checkout quote پیداست.

Canonical pricing model

```
CostObservation
+ LandedCostVersion
+ MarketPriceObservation
+ PricingRuleVersion
→ PriceCalculation
→ PriceApprovalDecision
→ PriceVersion
→ OfferPrice
→ CheckoutPriceSnapshot
```

حقایق را Client unitPrice این که نه کنند، ذخیره rule/version/effective time با را Price Snapshot بای Order و Cart کنند. تلقی

03 — Cart, Checkout, and AI Insights

Cart implementation

دارند: را زیر رفتارهای CartService و CartController

- create/get cart by user or session!
- get by cart ID!
- add/update/remove item!
- coupon attach/remove!
- guest-to-user merge!
- clear and expiry cleanup.

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Cart security and commercial integrity gaps

- شوند. می پذیرفته request از cartId و sessionId، userId،
- ندارند. آشکار ownership check by cart ID get/update/delete
- پذیرد. می Client از را attributes و providerId، currency، unitPrice، name، SKU، فیلهای Cart item
- نشد. دیده از قبل add از قبل Catalog/Offer/Inventory lookup
- شوند. نمی availability validate و channel eligibility، maximum quantity، MOQ
- انجام CouponService توسط بَعْدًا واقعی validation گویدمی صریحًا comment شود؛ می cart attach به فقط Coupon
- شود. می
- شود. می انجام Caller ارسالی sessionId/sessionId یا merge guest

Checkout implementation

CheckoutController:

- checkout start!
- cart-to-order conversion!
- checkout validation!
- order verification!
- payment confirmation!
- AI insights.

نمی انجام Catalog/Offer با واقعی comparison کنند. می منتقل Order Item به مستقیماً را Cart های قیمت checkout اما کنند. override را آن Order layer در TenantContext مگر شود، می Order وارد Checkout request از نیز tenantId شود.

confirm-payment، signed provider callback، idempotency، amount/currency reconciliation، Payment Intent ownership یا Order مستقیماً confirm-payment

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Simulated AI

کنند. می مغرفی AI decisions را reference implementation that simulates AI analysis صریحًا کد

Fraud score، ETA، payment suggestion، warehouse routing و upsell با ruleهای و hard-coded می ساخته شوند؛ برای شوند. می انتخاب فرضی های carrier و wh-special و wh-main های warehouse نمونه

Classification:

- AI checkout advice: SIMULATED_REFERENCE_IMPLEMENTATION
- persisted AI decision: PLANNED_NOT_IMPLEMENTED؛ کند. می log را آن فقط کد

Required checkout contract

Checkout: باید:

1. actor/cart ownership را verify کنند؛
 2. Offer و Price Version را resolve دوباره کنند؛
 3. availability و reservation را اجرا کنند؛
 4. shipping/tax/discount را authoritative محاسبه کنند؛
 5. quote expiry و idempotency داشته باشند؛
 6. Payment Intent بسازد؛
 7. Order Snapshot immutable ایجاد کنند؛
 8. AI output را advisory و traceable نگه دارد.
-

04 — Customer Order and Workflow

Implemented order model

OrderEntity دارای:

- Order, Payment و Fulfilment status؛
- tenantId؛
- monetary totals و currency؛
- item/address snapshots؛
- payment/shipping references؛
- history؛
- @Version optimistic locking؛
- transition methods برای approve/process/ship/deliver/cancel/hold/refund/return.

OrderApplicationService CRUD, workflow, analytics, import و editing را facade می کنند.

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Positive controls

- several state transitions validate current state؛
- entity uses enums؛
- optimistic version exists؛
- per-ID reads and workflow commands use TenantContext-aware lookup when context exists؛
- cache keys often include tenant key؛
- order history entity exists.

Critical findings

1. OrderCrudService.list() در branch repository.findAll() tenant-scoped list می گذارد؛
2. bulk approve/cancel/refund از findAllById بدون tenant filter استفاده می کنند.
3. system/internal context fail-open عمل می کند.
4. Order Items request را snapshot و قیمت پذیرند.
5. recordTransition status، همان status را فعلی و from و to خواند؛
6. markPaid state, amount, currency, transaction uniqueness و provider signature guard ندارد.
7. hold از status قابل اعمال است.
8. local Spring Event broker/outbox وجود ندارد؛
9. OrderWorkflowOrchestrator مراحل Inventory Reservation, Payment Capture, Fulfilment, Settlement و Returns را گرفته است.
10. auto-approval threshold 1000، currency normalization و risk policy بدون است.

Correct truth boundaries

اما: دارد؛ نگه را تجاری تعهد باید Order

- Payment outcome به Payment Domain متعلق است.
- Inventory reservation به Inventory متعلق است.
- Pick/Pack به Warehouse متعلق است.
- Shipment/Delivery observation به Shipping/Carrier integration متعلق است.
- Settlement به Finance/Marketplace settlement owner متعلق است.

ها. آن جایگزین نه باشد، facts این projection باید Order status

05 — Allocation, Fulfilment, Pick, Pack, and Last Mile

Allocation

Inventory Reservation در Wave 02 وجود دارد اما اتصال Order به Inventory در OrderWorkflowOrchestrator فقط comment است:

- validate products;
- reserve inventory;
- pre-authorize/capture payment;
- create fulfilment order.

Order-driven Allocation برابر با PLANNED_NOT_IMPLEMENTED است.

Legacy Fulfilment

Warehouse FulfillmentService:

- deprecated و forRemoval است;
- Sales Order list/get/update status دارد;
- get/update by ID tenant scope ندارند؛ آشکار است.
- status string آزاد است.

Classification: DEPRECATED_IMPLEMENTATION.

Pick and Pack

کنند: می مدیریت را Subjects این واقعاً Controller و PickPackService

- Pick Wave;
- Pick Task;
- Pack Task;
- assign/start/complete/cancel;
- picked quantity و short pick;
- tracking number, shipping label, weight, volume و box count.

Classification: IMPLEMENTED_WITH_GAP.

Pick/Pack gaps

- tenantId از request می گرفته شود.
- get/update operations by ID tenant-scoped نیستند.
- status و transition string محدود/ناهمگون است.
- confirmPick over-pick را نمی منع کند؛ quantity برابر با requested, PICKED می شود.
- Inventory reservation consumption/decrement نشده.
- wave/task creation به Order/Inventory entitlement متصل نیست.
- Pack Task order/shipment references را verify نمی کند.
- Event/outbox برای Picked/Packed پیدا نمی شود.
- Dispatch/Handover از Pack completion و جدا canonical نشده است.

Last-Mile Optimizer

LastMileCarrierOptimizer:

- Carrier profile های DHL, FedEx, UPS و China Post در memory seed می کند؛
- static rate table دارد؛

06 — Returns, Inspection, Disputes, and Refund

Returns implementation

دارند: را زیر رفتارهای ReturnsController و ReturnsService

- create return/RMA؛
- create return lines؛
- receive at warehouse؛
- inspect received items؛
- create return-to-stock decision؛
- dispute claim, evidence, rebuttal, decision و appeal workspace.

Classification: IMPLEMENTED_WITH_CRITICAL_GAPS.

Strong structural evidence

- هست. جدا entities Return, Return Item, Receipt, Inspection و Return-to-Stock Decision
- شود. می ذخیره Receipt و Return روی tenant/company
- شود. می نگاشت RESTOCK/HOLD/DAMAGED/SCRAP به classification
- دارد. appeal و financial impact, liability, evidence/timeline workspace dispute

Critical returns findings

1. می authorizationStatus = APPROVED فوراً create در Return هر
2. actor نه شوند، می تولید تصادفی UUID با executedBy و inspectedBy, receivedBy, createdBy
3. شود. نمی SKU ownership/eligibility verify و customer/supplier, source order line
4. ن دارند. آشکار tenant/company guard و inspect by receiptId و receive by returnId
5. کنند. می received برابر scan/discrepancy بدون را requested quantity تمام
6. ن دارد. authority/calibration/evidence و شود می پذیرفته request از inspection result/classification
7. است. شده موکول آینده به comment در inventory posting شود؛ می log فقط RESTOCK
8. نیست. متصل Payment Domain به Refund execution
9. خوانند. می tenant predicate بدون ID را record مسیری چند در appeal و decision, rebuttal, evidence, dispute detail
10. دارد. in-memory tenant filtering و () findAll() برای ReturnId workspace
11. شود. می ذخیره JSON mutable در evidence/timeline
12. نیست. وصل payout/refund execution به dispute financial decision

Canonical return flow

ReturnRequest
→ EligibilityDecision
→ ReturnAuthorization
→ InboundShipment/Receipt
→ Item-Level Inspection Evidence
→ Disposition Decision
→ Inventory Posting or Scrap/Repair
→ Refund/Replacement Intent
→ Payment Outcome
→ Resolution and Appeal

07 — Settlement, Reconciliation, and Payout

Competing settlement owners

دارد: وجود اجرای Settlement Service دو

1. Marketplace Settlement:
 - cycles!
 - transactions!
 - commission/discount aggregation!
 - reconciliation!
 - dashboard!
 - scheduler.
2. Payment Settlement:
 - batch های Payment Capture کردن!
 - gross/fee/net calculation!
 - batch به capture اتصال.

ownership: وضعیت

CONFLICT_REQUIRES_CANONICALIZATION

Marketplace Settlement

ملاحظات: نقاط

- state transition map!
- cycle calculation!
- audit و metrics!
- reconciliation records!
- auto-generation با overlap check!
- tenant-aware list endpoints.

Classification: IMPLEMENTED_WITH_GAP.

Material gaps

- tenant/company از request می create وارد شوند.
- get/update/calculate cycle by ID tenant guard ندارد. آشکار
- addTransaction request tenant/company با cycle یکسان باشد.
- transaction reference authenticity و idempotency دیده نشد.
- Dashboard totalCycles را global cycleRepository.count() می سازد و cross-tenant count leak.
- dashboard monetary summaries صرفه hard-code شده اند.
- status SETTLED state update فقط bank/payout provider execution پیدا نشد.
- reconciliation actualAmount و resolvedBy از request می پذیرفته
- LLM settlement disabled است.
- multi-currency aggregation normalization دیده نشد.

Payment Settlement

Payment service capture های Settlement Batch می سازد و fee rate را اعمال می کند، می جمع globally را نشده تسویه های

Classification: IMPLEMENTED_WITH_GAP.

Material gaps

- list و run settlement tenant/company scope ندارند.

- نیست. payout destination یا seller, tenant, currency اساس بر capture grouping
- شوند. می جمع batch یک در CAPTURED records تمام
- نشد. دیه failure recovery و reconciliation, retry, provider transfer, payout execution
- است. نشده روشن Marketplace Settlement با overlap

Canonical boundary

Payment money movement و provider facts را مالک باشد. Marketplace باید seller ledger, commission, entitlement باشد. payout success معنی به خوددببه نباید Settlement approval باشد. مالک را reconciliation و

09 — AI Evaluation and Governed Learning

Implemented evaluation capability

دارد: را رفتارها این AI Orchestrator

- submit human evaluation برای Agent Run؛
- recent evaluations by tenant؛
- evaluations by run؛
- use-case stats؛
- autonomy-level recommendation.

Rolling averages و recommendation rule است. tags و rating، accuracy، safety، usefulness، comment، corrections و شامل Evaluation میشوند. می اجرا نیز ها

Classification: IMPLEMENTED_WITH_GAP.

Positive evidence

- rating enum-like validation دارد.
- aggregate stats per tenant/use case می ذخیره شود.
- high/low safety and accuracy thresholds recommendation تولید کنند.
- minimum sample sizes برای escalation بخ ی است. شده گرفته نظر در ها

Critical evaluation findings

1. evaluatorUserId از request پذیرفته فقط و prefix human: می ذخیره شود.
2. Agent Run existence، completion، tenant ownership و use-case match verify نمی شود.
3. GET evaluations/{runId} header tenant را ول ی گ یردم ی Service فقط بر ف query runId اساس بر ف
4. accuracy/safety/usefulness score range validate نمی شود.
5. duplicate evaluation ی evaluator conflict policy دیه نشد.
6. evaluation independence، reviewer role، blind review و conflict-of-interest وجود ندارد.
7. corrections/tags می ذخیره strings/JSON صورت به
8. recommendation فقط heuristic و است policy change approval/خودکار governance کامل ندارد.
9. default current autonomy level 3 برابر است، profile evidence بدون ح تی است.
10. evaluation quality، dataset version، model version، prompt/tool configuration و output artifact طور به snap- shot صریح طور به

Governed learning boundary

Evidence window، Recommendation باید ده. تغ ی ر را Autonomy ی Model مستقی م نباید Evaluation feedback، model/run version، reviewer authority، safety veto، approval و rollback همراه باشد.

10 — Cross-Domain Conflicts and Canonicalization

ID	Conflict	Canonical decision required
C-301	Product master در catalog.products و marketplace.product_catalog_entries شده تکرار	Product Master, Offer Projection و migration owner شود تعینی
C-302	Product, Marketplace Catalog, Target Price و Smart Pricing در قیمت دارد وجود	Price Version و Pricing authority واحد شود تعریف
C-303	Cart unit price از Client پذیردمی	Checkout بای Offer/Price Snapshot authoritative بسازد
C-304	confirm-payment Order را مستقیماً Paid کندمی	Payment provider outcome منبع تنه باشد truth
C-305	Order list و bulk operations tenant leak دارند	repository access، fail-closed tenant-scoped شوند
C-306	Order Saga comment فقط است	Inventory/Payment/Fulfilment orchestration و compensating actions شود ساخته
C-307	Warehouse Sales Order deprecated با Customer Order overlap دارد	Fulfilment Order boundary و migration شود تعینی
C-308	Pick/Pack به Inventory consumption نیست متصل	Reservation consumption و stock ledger contract پیاده شود
C-309	Delivered POD ثبت شودمی بدون	Delivery Observation و POD مستقل شود ایجاد
C-310	Return RESTOCK log فقط است	Return disposition به Inventory Posting متصل شود
C-311	Refund در Order/Return با Payment execution جداست	Refund Intent و Provider Outcome canonical شود
C-312	Marketplace و Payment دو هر Settlement owner هستند	Seller ledger و money movement ownership شود جدا
C-313	Sustainability Goal دو در Service وجود دارد	ESG/Carbon owner و projection، شوند تعینی
C-314	AI evaluation run lookup tenant-scoped نیست	Evaluation identity و run ownership شود گیری سخت
C-315	local/in-memory/static engines معرفیه AI/Carrier behavior عنوان به شوندمی	Simulation label و production provider boundary اجباری شود

11 — Implementation Status Matrix

Area	Narrow verified capability	Classification
Product Catalog Service	CRUD/status/resell/similar	IMPLEMENTED_WITH_GAP
Marketplace Product Catalog	tenant-aware table and CRUD	IMPLEMENTED_WITH_GAP
Canonical Marketplace Offer	Not found as independent aggregate	PLANNED_NOT_IMPLEMENTED
Target Pricing	Policy CRUD and lookup	IMPLEMENTED_WITH_GAP
Smart Pricing	Rules, landed cost, market snapshots, calculation	IMPLEMENTED_WITH_GAP
Price Version / Checkout Quote	No immutable accepted version	PLANNED_NOT_IMPLEMENTED
Cart	CRUD, merge, coupon attachment	IMPLEMENTED_WITH_CRITICAL_GAPS
Checkout	Cart-to-order and validation	IMPLEMENTED_WITH_CRITICAL_GAPS
Checkout AI	Hard-coded local rules	SIMULATED_REFERENCE_IMPLEMENTATION
Customer Order	Entity, workflow, history, optimistic lock	IMPLEMENTED_WITH_CRITICAL_GAPS
Order Saga	Comments only for cross-service steps	PLANNED_NOT_IMPLEMENTED
Allocation	Inventory reservation exists, order orchestration absent	PARTIAL_IMPLEMENTATION
Legacy Fulfilment Sales Order	Deprecated CRUD/status	DEPRECATED_IMPLEMENTATION
Pick/Pack	Waves/tasks/confirm/pack/verify	IMPLEMENTED_WITH_GAP
Last-Mile Optimizer	Static in-memory rates and profiles	SIMULATED_REFERENCE_IMPLEMENTATION
Proof of Delivery	Not found	PLANNED_NOT_IMPLEMENTED
Returns/RMA	Return, receipt, inspection, disposition	IMPLEMENTED_WITH_CRITICAL_GAPS
Disputes	Evidence, rebuttal, decision, appeal	IMPLEMENTED_WITH_GAP
Marketplace Settlement	Cycles, transactions, calculation, reconciliation	IMPLEMENTED_WITH_GAP
Payment Settlement	Capture batching	IMPLEMENTED_WITH_GAP
Payout Execution	Not verified	PLANNED_NOT_IMPLEMENTED
ESG Goals	CRUD, progress, events	IMPLEMENTED_WITH_GAP
Verified ESG Metrics	Not found	PLANNED_NOT_IMPLEMENTED
AI Evaluation	Feedback, stats, recommendations	IMPLEMENTED_WITH_GAP

12 — Wave 03 Issue Register

ID	Severity	Area	Finding	Required action
W3-001	Critical	Catalog	ProductEntity has no tenant/company field	Add canonical tenant ownership or explicitly global product governance.
W3-002	Critical	Catalog	Product update can set status directly	Remove status from generic update and enforce transition/role policy.
W3-003	High	Catalog	Product status update lacks visible authorization and transition matrix	Require actor role, allowed transitions, reason and version.
W3-004	High	Catalog	Product has no optimistic version	Add @Version and conflict handling.
W3-005	Critical	Catalog	Resell-from-purchase does not verify source ownership or stock eligibility	Validate order ownership, receipt, quality and resale rights.
W3-006	Critical	Catalog	Three product/catalog representations compete	Canonicalize Product Master, Marketplace projection and Offer.
W3-007	High	Marketplace Catalog	Tenant/company and verification flags come from request	Derive identity and restrict verified/featured changes.
W3-008	Critical	Marketplace Catalog	Get/update/delete by ID are not visibly tenant-scoped	Use tenant-qualified repository methods.
W3-009	High	Marketplace Catalog	Hard delete removes catalog evidence	Use archive/supersession and retention policy.
W3-010	Critical	Offer	No independent Marketplace Offer aggregate found	Implement seller/product/price/availability/terms snapshot.
W3-011	Critical	Pricing	Pricing authority is split across Product, Marketplace Catalog, Target Price and Smart Pricing	Create canonical PriceVersion ownership.
W3-012	Critical	Pricing	Tenant IDs are accepted from request/query	Use trusted tenant context.
W3-013	High	Pricing	Rule get/update/delete by ID are unscoped	Use tenant-qualified lookup.
W3-014	High	Pricing	Category and warehouse scopes match all rules	Implement real category/warehouse matching.
W3-015	High	Pricing	Calculation response currency is hard-coded USD	Preserve and validate currency.
W3-016	High	Pricing	Manual adjustments have no approval/limit evidence	Add policy, authorization and audit.
W3-017	High	Pricing	Hard-coded volume tiers exist in multiple services	Govern pricing rule versions centrally.

ID	Severity	Area	Finding	Required action
W3-018	Critical	Cart	Cart ownership relies on request user/session/cart IDs	Bind carts to authenticated principal/session token.
W3-019	Critical	Cart	Client supplies SKU, unitPrice, provider and currency	Resolve authoritative offer server-side.
W3-020	High	Cart	Coupon code is stored without validation	Integrate governed promotion service.
W3-021	High	Cart	Quantity and availability policy is absent	Validate MOQ, max, inventory and channel rules.
W3-022	Critical	Checkout	Checkout converts client-priced cart directly to Order	Reprice and snapshot offer before order creation.
W3-023	Critical	Checkout	confirm-payment can directly mark Order paid	Require signed provider outcome, amount/currency match and idempotency.
W3-024	High	Checkout	Price validation is only zero-price warning	Compare PriceVersion and quote expiry.
W3-025	High	Checkout AI	AI insights are simulated but exposed as AI-powered	Label simulation and separate production adapter.
W3-026	High	Checkout AI	AI decisions are logged but not governed/persisted	Store advisory decision provenance or remove claim.
W3-027	Critical	Order	Tenant-scoped list executes repository. findAll	Fix repository call and add isolation test.
W3-028	Critical	Order	Bulk approve/cancel/refund use unscoped findAllById	Use tenant-scoped bulk lookup.
W3-029	High	Order	System context is fail-open	Require explicit privileged system identity.
W3-030	High	Order	History transition records current status for both from/to	Use captured old status and test history.
W3-031	Critical	Order	markPaid lacks payment outcome guards	Move payment truth to Payment service.
W3-032	High	Order	Auto-approval threshold ignores currency and risk	Use normalized amount and policy engine.
W3-033	Critical	Order Saga	Inventory/payment/fulfilment steps are comments only	Implement durable saga and compensation.
W3-034	High	Order Events	Only local Spring events are published	Add durable outbox/broker.
W3-035	Critical	Fulfilment	Deprecated SalesOrder get/update is unscoped	Remove or tenant-scope during migration.
W3-036	Critical	PickPack	Task/wave IDs are not tenant-scoped	Use tenant/warehouse-qualified lookup.
W3-037	High	PickPack	Over-pick is accepted as PICKED	Reject or explicitly handle excess quantity.
W3-038	Critical	PickPack	No inventory reservation	Integrate Inventory ledger and idempotency.
W3-039	High	PickPack	consumption found Pack references are not validated	Verify order/shipment/warehouse relation.
W3-040	High	Last Mile	Carrier rates and profiles are static in memory	Use versioned provider quotes and live adapters.
W3-041	Critical	Delivery	Order can become DELIVERED without POD	Require qualified carrier/POD observation.

ID	Severity	Area	Finding	Required action
W3-042	Critical	Returns	Returns are auto-approved on creation	Implement eligibility and authorization decision.
W3-043	Critical	Returns	Actors are random UUIDs	Use authenticated actor identity.
W3-044	Critical	Returns	Receive/inspect paths lack visible tenant guard	Tenant-scope return and receipt operations.
W3-045	High	Returns	Receive sets all requested quantity as received	Capture item/package discrepancies.
W3-046	Critical	Returns	RESTOCK only logs; Inventory is not updated	Execute idempotent Inventory posting.
W3-047	High	Disputes	Detail/evidence/rebuttal/decision paths query case by ID without tenant predicate	Enforce case ownership.
W3-048	High	Disputes	Evidence and timeline are mutable JSON	Use append-only evidence records.
W3-049	Critical	Settlement	Marketplace and Payment settlement ownership overlaps	Separate seller ledger from money movement.
W3-050	Critical	Settlement	Marketplace cycle/transaction ID operations are unscoped	Tenant-scope every operation.
W3-051	Critical	Settlement	Transaction tenant/company may differ from cycle	Derive scope from cycle and validate reference.
W3-052	High	Settlement	Dashboard totalCycles is global	Use tenant-scoped count.
W3-053	High	Settlement	Dashboard monetary values are zero placeholders	Calculate real tenant metrics.
W3-054	Critical	Payment Settlement	All captured payments can be grouped globally	Partition by tenant, seller, currency and payout destination.
W3-055	Critical	Payout	SETTLED state has no verified provider transfer	Separate approval from payout outcome.
W3-056	Critical	ESG	Goal ID operations and getAll are unscoped	Use trusted tenant context and role policy.
W3-057	High	ESG	Progress accepts self-reported currentValue without evidence	Attach measurement source and methodology.
W3-058	High	ESG	Progress formula may compare percentage to absolute target	Define target semantics and validate units.
W3-059	High	ESG	Science-based/SBTi status is self-asserted	Store external assurance evidence.
W3-060	High	ESG	Kafka failure is swallowed and payload is minimal	Use outbox and governed event envelope.
W3-061	Critical	ESG	Duplicate Sustainability Goal models exist	Canonicalize Compliance and specialized Carbon ownership.
W3-062	Critical	AI Evaluation	Evaluations-by-run ignore tenant header	Query by tenant and run.
W3-063	Critical	AI Evaluation	Evaluator identity is client-supplied	Use authenticated reviewer identity and role.
W3-064	High	AI Evaluation	Agent run existence and tenant ownership are not verified	Link evaluation to immutable run record.
W3-065	High	AI Evaluation	Scores have no range validation	Validate 0-100 and missing-value policy.

ID	Severity	Area	Finding	Required action
W3-066	High	AI Evaluation	Duplicate/conflicting evaluations are unrestricted	Define reviewer and revision policy.
W3-067	High	AI Evaluation	Model/prompt/tool/dataset versions are not snapshotted	Bind evaluation to reproducible run context.
W3-068	High	Cross-domain	No durable outbox found across reviewed workflows	Standardize outbox/inbox and replay controls.

13 — Wave 03 Traceability Matrix

Evidence ID	Repository path	Blob SHA	Classification
CATALOG-PRODUCT-CONTROLLER	backend/domain/commerce/product-catalog-service/src/main/java/com/allinb2c/commerce/productcatalog/api/ProductController.java	5d1153964968cc4a51674833c8006f6651f25677	IMPLEMENTED WITH_GAP
CATALOG-PRODUCT-SERVICE	backend/domain/commerce/product-catalog-service/src/main/java/com/allinb2c/commerce/productcatalog/service/ProductService.java	54b0cbc821d6480be31585254405a1f6007e06b7a	IMPLEMENTED WITH_GAP
CATALOG-PRODUCT-ENTITY	backend/domain/commerce/product-catalog-service/src/main/java/com/allinb2c/commerce/productcatalog/entity/ProductEntity.java	bad889f79bbcb281e5f0afae11b6018b527b95	IMPLEMENTED WITH_GAP
MARKETPLACE-CATALOG-MAPPING	backend/domain/commerce/marketplace-service/src/main/java/com/allinb2c/commerce/marketplace/catalog/CatalogProductEntity.java	5a78fddadc333c2b49458e3d8c01560c658e0100	REQUIRES_CANONICALIZATION
MARKETPLACE-ECOSYSTEM-DDL	backend/domain/commerce/marketplace-service/src/main/resources/db/migration/V8__phase_h_marketplace_ecosystem.sql	e3f04355db77b411a03d77ff117f7126d110677	IMPLEMENTED
MARKETPLACE-CATALOG-CONTROLLER	backend/domain/commerce/marketplace-service/src/main/java/com/allinb2c/commerce/marketplace/controller/ProductCatalogController.java	6b7a2675f1283e2fece8cala997b4c45d70126	IMPLEMENTED WITH_GAP
MARKETPLACE-CATALOG-SERVICE	backend/domain/commerce/marketplace-service/src/main/java/com/allinb2c/commerce/marketplace/service/ProductCatalogService.java	1d5c280c0e5086c5fd7ed34e9411b6018b527b95	IMPLEMENTED WITH_GAP
TARGET-PRICE-CONTROLLER	backend/domain/commerce/pricing-service/src/main/java/com/allinb2c/commerce/pricing/controller/TargetPriceController.java	4dd2bf14a963a27d9a4276df4d5b7036116d56b	IMPLEMENTED WITH_GAP
TARGET-PRICE-SERVICE	backend/domain/commerce/pricing-service/src/main/java/com/allinb2c/commerce/pricing/service/TargetPriceService.java	8e1dfde40c29a53dcae6cd4e710915f621f95d19	IMPLEMENTED WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
SMART-PRICING-CONTROLLER	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ controller/ SmartPricingController. java	20079cc399fcc65d9512715c50b19f1e839d5045	IMPLEMENTED WITH GAP
SMART-PRICING-SERVICE	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ service/ SmartPricingService.java	8be4df025c3836f5395284415149268545f86961	IMPLEMENTED WITH GAP
CART-CONTROLLER	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/api/ CartController.java	2f1f550db8296746b41f6e87ea1f489961f5d0013	IMPLEMENTED WITH CRITICAL_GAPS
CART-SERVICE	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ CartService.java	9f7194377156c248973ebba6a150764e5f05d5060	IMPLEMENTED WITH CRITICAL_GAPS
CHECKOUT-CONTROLLER	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/api/ CheckoutController.java	77cf28681bd107a6cc022793812041e8806f08365	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-APPLICATION	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderApplicationService. java	7fba54d4dfb94afcc8d2f57511011046505df1600	IMPLEMENTED WITH GAP
ORDER-ENTITY	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/entity/ OrderEntity.java	1f3efa66e3830dc2c7e85866fa1808f08a060b517	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-CRUD	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderCrudService.java	df6182d7681f495b4ba7915ad1401c281f5d6735	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-WORKFLOW	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderWorkflowService. java	925e9a08b779fa5bb3fae3e2f1040414f9c28e98f	IMPLEMENTED WITH CRITICAL_GAPS
ORDER-EVENT-PUBLISHER	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/event/ OrderEventPublisher.java	2e845993a7a7f582bf29c5f5e6c0414f9c28e98f	LOCAL EVENT ONLY
ORDER-ORCHESTRATOR	backend/domain/commerce/ order-service/src/main/ java/com/allinb2c/ commerce/order/service/ OrderWorkflowOrchestra- tor.java	85b7b03ac88825db7ad2862e2f04b1ed63d7c15	PUBLISHED BUT NOT IMPLEMENTED
LEGACY-FULFILMENT	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ service/ FulfillmentService.java	9f7906ea15fb5b7717bc03dbcb4e226c07f00b084	DEFERRED TO IMPLEMENTATION

Evidence ID	Repository path	Blob SHA	Classification
PICK-PACK-SERVICE	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ service/PickPackService. java	b34fe872f1b365d1fe81bd42ea109f22621eb0873	TEMPORARY_WITH_GAP
PICK-PACK-CONTROLLER	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ controller/ PickPackController.java	19619f06c31753480139e2abb0106941e8390611	TEMPORARY_WITH_GAP
LAST-MILE-OPTIMIZER	backend/domain/ fulfillment/warehouse- service/src/main/java/ com/allinb2c/ fulfillment/warehouse/ routing/LastMileCar- rierOptimizer.java	b937fc261f5f0b36017526b4e5007a3f1d8c77	REFERENCE_IMPLEMENTATION
RETURNS-CONTROLLER	backend/domain/commerce/ returns-service/src/ main/java/com/allinb2c/ commerce/returns/ controller/ ReturnsController.java	1dfa9a38b5aa5ea897983f57a8671e5b93e06969	CRITICAL_GAPS
RETURNS-SERVICE	backend/domain/commerce/ returns-service/src/ main/java/com/allinb2c/ commerce/returns/ service/ReturnsService. java	56f7daeb6050444bcf661de674504e97899b4478	CRITICAL_GAPS
RETURNS-WORKSPACE	backend/domain/commerce/ returns-service/src/ main/java/com/allinb2c/ commerce/returns/ service/ReturnsResolu- tionWorkspaceService. java	5e2054b768d6714eb97a6fade818c2f1243e97628	TEMPORARY_WITH_GAP
MARKETPLACE-SETTLEMENT- CONTROLLER	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ controller/ SettlementController. java	b7c07f242f699f22817ef1fa8a10f2180494e99a41	TEMPORARY_WITH_GAP
MARKETPLACE-SETTLEMENT- SERVICE	backend/domain/commerce/ marketplace-service/src/ main/java/com/allinb2c/ commerce/marketplace/ service/ SettlementService.java	e1a9c2bc7fa3d534d6473c8fd8c1f1560f03616	TEMPORARY_WITH_GAP
PAYMENT-SETTLEMENT- SERVICE	backend/domain/commerce/ payment-service/src/ main/java/com/allinb2c/ commerce/payment/ service/ SettlementService.java	d22c93a39b7bef2beabc33a5e02b489860f62b21	TEMPORARY_WITH_GAP
ESG-GOAL-CONTROLLER	backend/domain/ engagement/compliance- service/src/main/java/ com/allinb2c/engagement/ compliance/ carbontracking/api/ SustainabilityGoalCon- troller.java	6adbd69c94842195498fad0a7101f50c990e30	TEMPORARY_WITH_GAP

Evidence ID	Repository path	Blob SHA	Classification
ESG-GOAL-SERVICE	backend/domain/engagement/compliance-service/src/main/java/com/allinb2c/engagement/compliance/carbontracking/service/SustainabilityGoalService.java	f57f361dd603d06487a7584fa1e07f06d40943a	UNCLASSIFIED - WITH GAP
AI-EVALUATION-CONTROLLER	backend/domain/intelligence/ai-orchestrator/src/main/java/com/allinb2c/intelligence/orchestrator/evaluation/EvaluationController.java	4477c54867d7113c70c7f93cd17035070c4e9a28	UNCLASSIFIED - WITH GAP
AI-EVALUATION-SERVICE	backend/domain/intelligence/ai-orchestrator/src/main/java/com/allinb2c/intelligence/orchestrator/evaluation/EvaluationService.java	66445f266f5aae7347cf1bd1c610945f8787d9041a	UNCLASSIFIED - WITH GAP

14 — Remediation and Verification Plan

Priority 0 — Security and Commercial Integrity

1. Fix Order list() tenant leak and all bulk operation leaks.
2. Scope Cart, Marketplace Catalog, Pricing, Fulfilment, Returns, Settlement, ESG and AI Evaluation ID lookups to trusted tenant context.
3. Stop accepting authoritative price, user, payment and verification state from Client input.
4. Disable or protect direct confirm-payment, direct product approval and self-verified catalog fields.
5. Correct Order history from/to status recording.
6. Replace random actor UUIDs in Returns with authenticated actor identity.
7. Require POD before Order becomes Delivered.

Priority 1 — Canonical Ownership

1. Product Master versus Marketplace Catalog/Offer.
2. Price authority and immutable Price Version.
3. Customer Order versus Warehouse Fulfilment Order.
4. Marketplace settlement ledger versus Payment payout execution.
5. Compliance ESG versus specialized Carbon service.
6. Return disposition versus Inventory stock posting.

Priority 2 — Reliable Execution

1. Transactional outbox/inbox for Order, Pick/Pack, Returns, Settlement, ESG.
2. Order Saga with inventory reservation, payment authorization, fulfilment creation and compensation.
3. Idempotent Checkout, Payment confirmation, Return receive/inspect and Settlement transaction ingestion.
4. Live carrier adapter, quote expiry, booking, tracking webhook and POD.
5. Immutable evidence journals instead of mutable JSON timelines.
6. Multi-currency normalization and financial reconciliation.

Required verification

- tenant isolation tests including list/bulk/cache paths:
 - client price tampering E2E:
 - duplicate checkout/payment webhook tests:
 - order saga failure/compensation tests:
 - concurrent stock allocation and over-pick tests:
 - return quantity discrepancy and duplicate restock tests:
 - settlement cross-tenant/currency/provider reconciliation:
 - ESG methodology/version/assurance tests:
 - AI evaluation run ownership, score range and duplicate-review tests:
 - CI logs and migration upgrade evidence.
-

15 — Repository Evidence Wave 04 Plan

کند: بررسی را cross-cutting زیرساخت باید Wave 04

1. Identity, Authentication, Authorization و Tenant Isolation
2. Audit, Evidence Integrity و Non-Repudiation
3. Event Bus, Outbox/Inbox, Idempotency و Replay
4. API Gateway, Rate Limiting و Abuse Protection
5. Secrets, Key Management و Encryption
6. Observability, SLO, Incident و Disaster Recovery
7. Database ownership, migration safety و backup/restore
8. Data Privacy, Retention, Consent و Residency
9. CI/CD, Supply Chain Security و Deployment Evidence
10. Production Readiness and Go-Live Evidence

شود. اعلام production-certified نباید ت جاری Stage ۰۴ Wave بدون

Implementation and Evidence Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable code or migration supports the narrow claim
IMPLEMENTED_WITH_GAP	Executable behavior exists with material gaps
IMPLEMENTED_WITH_CRITICAL_GAPS	Executable behavior exists but security or truth-integrity defects block safe use
PARTIAL_IMPLEMENTATION	Only a subset of the capability is executable
DEPRECATED_IMPLEMENTATION	Executable but explicitly deprecated or pending removal
LOCAL_EVENT_ONLY	In-process event exists without durable broker/outbox evidence
SIMULATED_REFERENCE_IMPLEMENTATION	Static, in-memory, or rule simulation; not a production external integration
PLANNED_NOT_IMPLEMENTED	Intent/comments/contracts exist without executable end-to-end behavior
CONFLICT_REQUIRES_CANONICALIZATION	Competing owners or Systems of Record exist
REQUIRES_REPOSITORY_REVIEW	Evidence remains insufficient
TEST_FILE_DISCOVERED	Test source exists; execution result is unverified

GTOS Handbook — Batch 09

Repository Evidence Wave 03

کند:می بررسی Repository اجرای کد اساس بر را Customer-facing Commerce بخش بسته این

- Product Catalog and Marketplace Catalog
- Pricing and Price Calculation
- Cart and Checkout
- Customer Order
- Allocation, Pick, Pack, Fulfilment
- Last-Mile and Delivery Proof
- Returns, Inspection, Disputes
- Settlement and Reconciliation
- ESG / Sustainability Goals
- AI Evaluation and Governed Learning

Field	Value
Repository	chinair88-prog/allinb2c
Evidence snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
GitHub branch	gtos/handbook-repository-grounded-expansion-v2
Release	Batch 09 / Evidence Wave 03
Publication maturity	Code-grounded review artifact

کند.نمی اثبات را production certification یا Deployment, CI success, provider outcome بسته این

Batch 09 Release Report

Field	Value
Release	Repository Evidence Wave 03
Repository snapshot	14cc83805e7386c3d5cdec6ad191b1078fd502ff
GitHub branch	gtos/handbook-repository-grounded-expansion-v2
Scope	Catalog through AI Evaluation
Maturity	Review artifact, not production certification

Principal conclusions

- Product/Catalog and Pricing have multiple competing authorities.
 - Cart and Checkout trust Client-supplied identity and price.
 - Customer Order has a tenant leak in list and unscoped bulk workflows.
 - Cross-service Order Saga is not implemented.
 - Pick/Pack is executable but not connected to Inventory consumption.
 - Last-Mile is a static in-memory optimizer and POD is absent.
 - Returns creates useful records but auto-approves, invents actor IDs, and does not execute restock/refund.
 - Marketplace and Payment both implement Settlement with unclear ownership.
 - ESG Goal tracking is self-reported and duplicated across services.
 - AI Evaluation is executable but run ownership, reviewer identity, score bounds and reproducibility are incomplete.
- No area in this Wave is certified end-to-end complete.

Implementation and Evidence Status Vocabulary

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Integrity Report

- Generated: 2026-07-23T15:53:11.726851+00:00
- Snapshot: 14cc83805e7386c3d5cdec6ad191b1078fd502ff
- Markdown files before this report: 20
- CSV files: 3
- Source entries: 34
- Issue entries: 68
- Critical issues: 33
- High issues: 35
- Word count before this report: 10,906
- Empty Markdown files: 0

Claim-discipline checks

```
{  
  "catalog_conflict_recorded": true,  
  "client_price_risk_recorded": true,  
  "order_tenant_leak_recorded": true,  
  "simulated_ai_recorded": true,  
  "pod_not_claimed": true,  
  "settlement_conflict_recorded": true,  
  "esg_assurance_boundary": true,  
  "ai_run_tenant_gap_recorded": true,  
  "issue_register_present": true,  
  "source_registry_present": true  
}
```

Result

PASS

PASS means package integrity and claim discipline only. It does not certify the software.

GTOS Handbook — Batch 09

Repository Evidence Wave 03

کند:می بررسی Repository اجرای کد اساس بر را Customer-facing Commerce بخش بسته این

- Product Catalog and Marketplace Catalog
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- Cart and Checkout
- Customer Order
- Allocation, Pick, Pack, Fulfilment
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